

Chapter 18

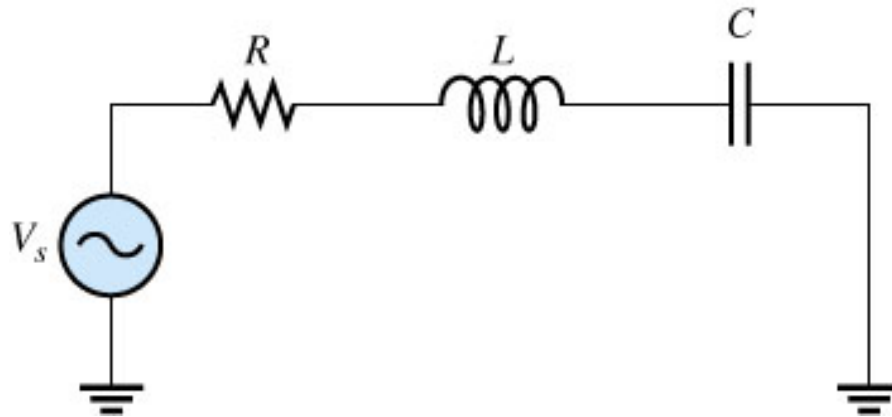
RLC Circuits and Resonance

Objectives

- Determine the impedance of a series RLC circuit
- Analyze series RLC circuits
- Analyze a circuit for resonance
- Determine the impedance of a parallel resonant circuit
- Analyze parallel RLC circuits
- Analyze a circuit for parallel resonance
- Analyze series-parallel RLC circuits
- Determine the bandwidth of resonant circuits

Impedance of Series RLC Circuits

- A series RLC circuit contains both inductance and capacitance.
- Since X_L and X_C have opposite effects on the circuit phase angle, the total reactance (X_{tot}) is less than either individual reactance.



Impedance of Series RLC Circuits

- When $X_L > X_C$, the circuit is predominantly inductive.
- When $X_C > X_L$, the circuit is predominantly capacitive.
- Total impedance for a series RLC circuit is:

$$\mathbf{Z} = R + jX_L - jX_C$$

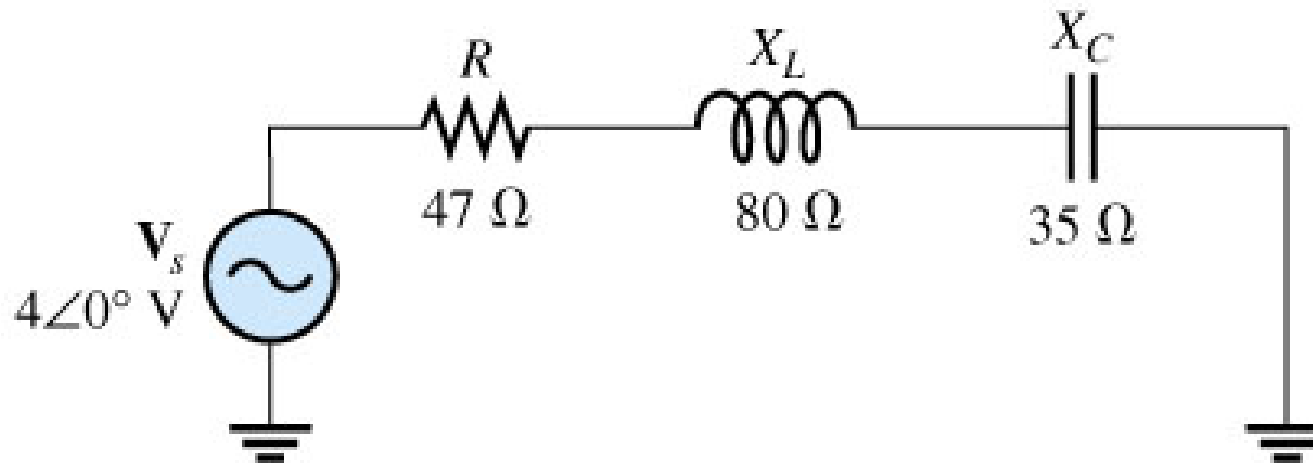
- In polar form:

$$\mathbf{Z} = \sqrt{R^2 + (X_L - X_C)^2} \angle \tan^{-1}(X_{\text{tot}}/R)$$

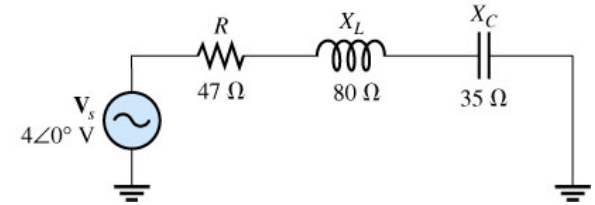
Series RLC Example 1

Find the following parameters if the source frequency is 5KHz:

- Circuit Impedance in polar and rectangular form.
- Is the circuit more inductive or more capacitive?
- Draw the impedance phasor diagram.
- The net reactance that will make the impedance magnitude equal to 100 ohms.



Series RLC Example 1 (cont.)



a). Circuit Impedance Z_T in rectangular and polar form:

- $Z_T = R + j(X_L - X_C) = 47\Omega + j(80\Omega - 35\Omega) = \underline{47\Omega + j45\Omega}$

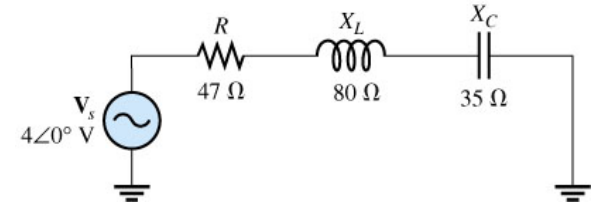
- $Z_T = \sqrt{R^2 + j(X_L - X_C)^2} = \sqrt{(47\Omega)^2 + j(45\Omega)^2} = \underline{65.1\Omega}$

- $\theta = \tan^{-1}\left(\frac{(X_L - X_C)}{R}\right) = \tan^{-1}\left(\frac{45\Omega}{47\Omega}\right) = \underline{43.8^\circ}$

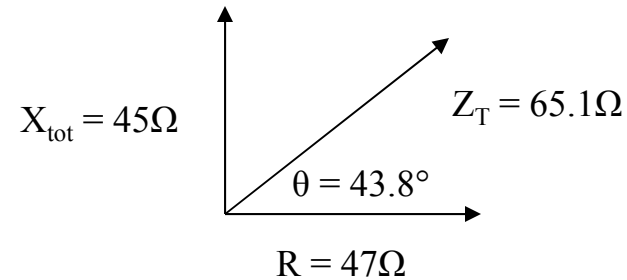
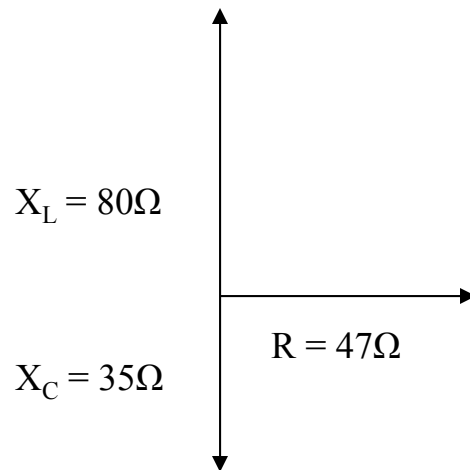
- $Z_T = \underline{65.1\Omega \angle 43.8^\circ}$

b). Since $X_L > X_C$, the circuit is more inductive.

Series RLC Example 1 (cont.)



c).



d).

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = 100 \, \Omega$$

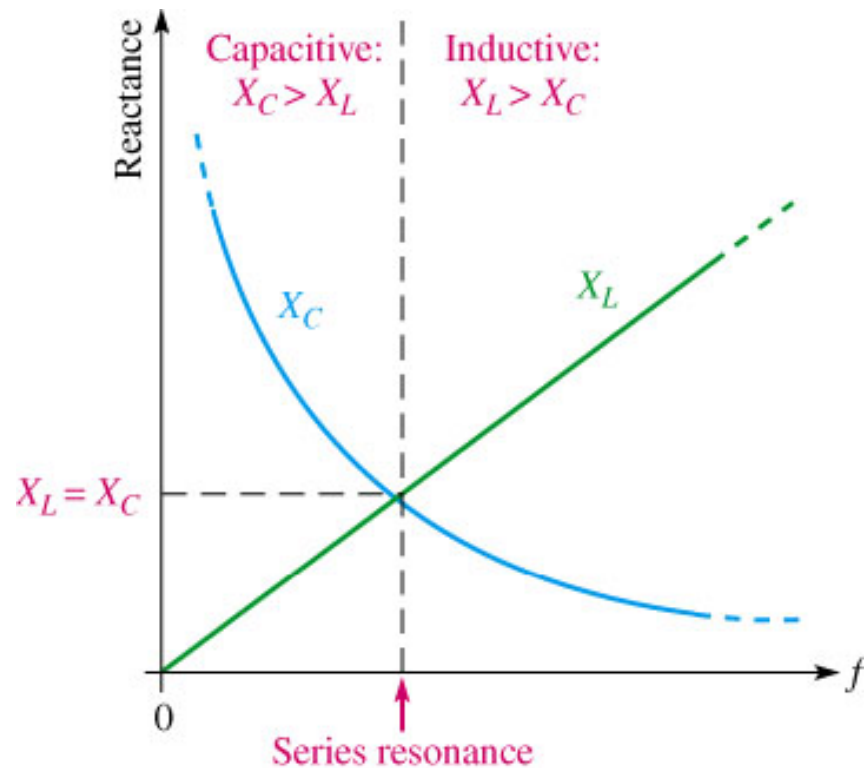
$$R^2 + (X_L - X_C)^2 = 100^2$$

$$(X_L - X_C)^2 = 100^2 - R^2$$

$$X_L - X_C = \sqrt{100^2 - R^2} = \sqrt{(100 \, \Omega)^2 - (47 \, \Omega)^2} = 88.3 \, \Omega$$

Analysis of Series RLC Circuits

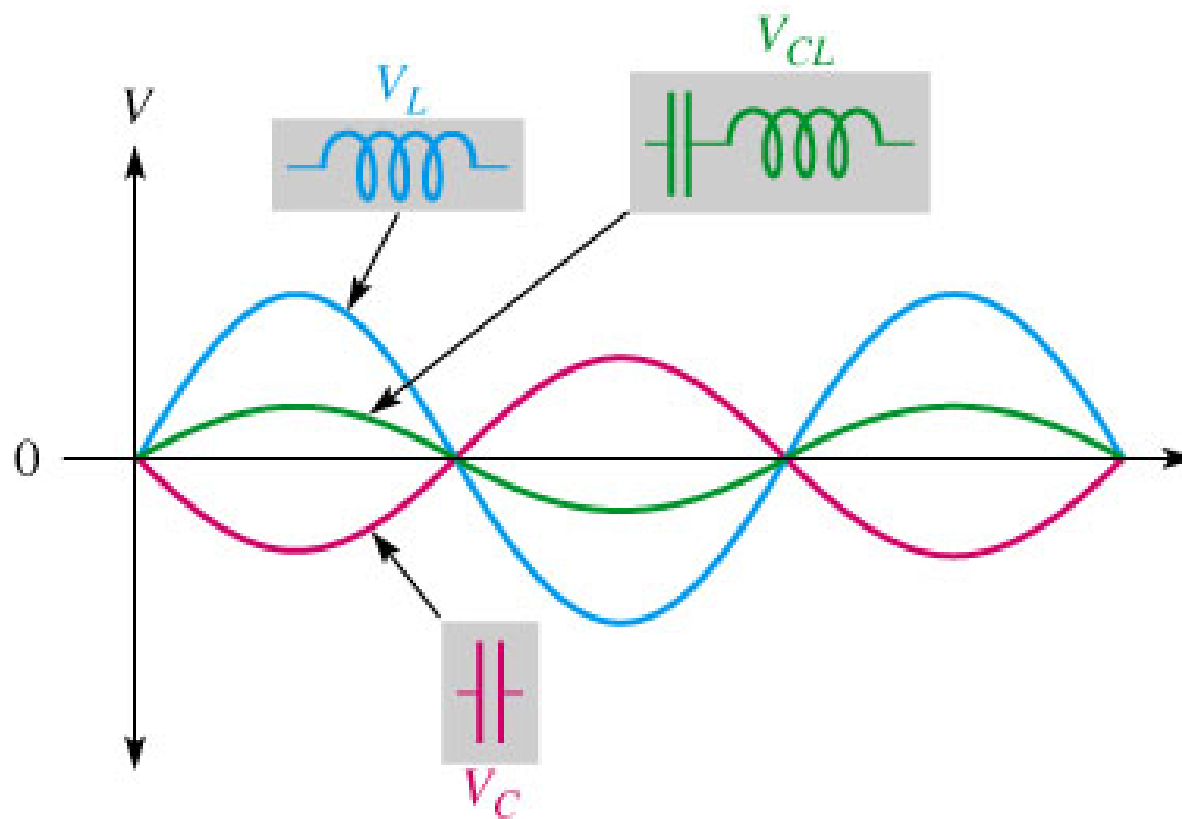
- A series RLC circuit is:
- Capacitive for $X_C > X_L$
- Inductive for $X_L > X_C$
- Resonant for $X_C = X_L$
- At resonance $Z_r = R$



Voltage Across the Series Combination of L and C

- In a series RLC circuit, the capacitor voltage and the inductor voltage are always 180° out of phase with each other.
- Because they are 180° out of phase, V_C and V_L subtract from each other.
- The voltage across L and C combined is always less than the larger individual voltage across either element.

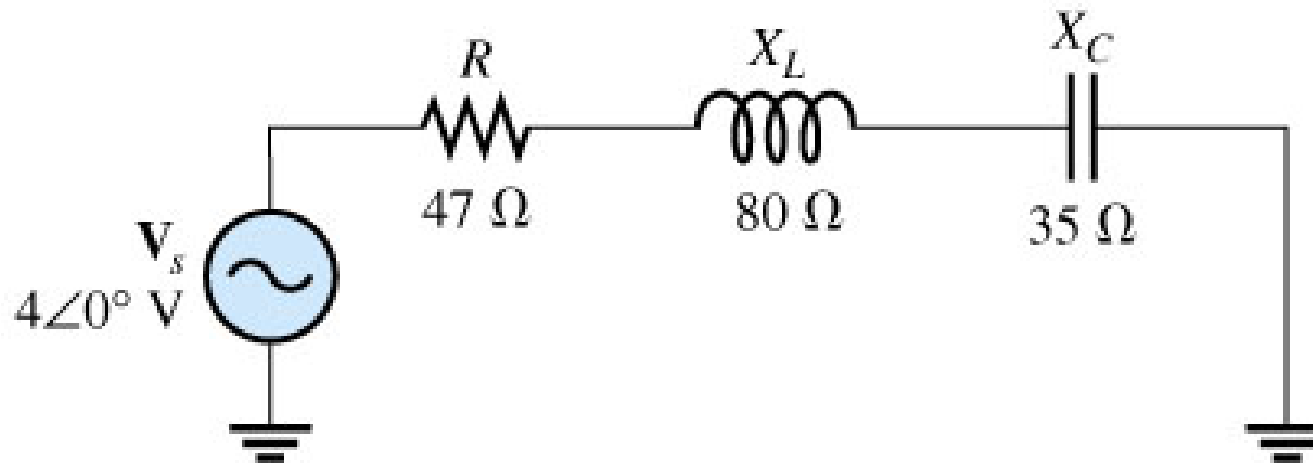
Figure 18--6 V_{CL} is the algebraic sum of V_L and V_C . Because of the phase relationship, V_L and V_C effectively subtract.



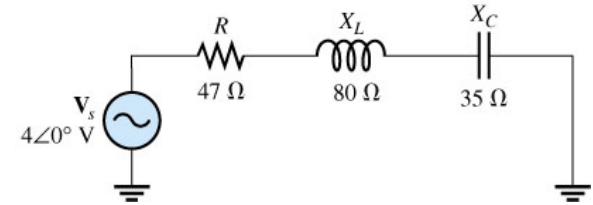
Series RLC Example 2

Find the following parameters if the source frequency is 5KHz:

- I_{tot} , V_R , V_L , and V_C in polar form.
- Draw the voltage phasor diagram.
- P_{true} , P_r , P_a , P_F , and power phasor diagram.



Series RLC Example 2 (cont.)



a). $\mathbf{Z_T = R + jX_L - jX_C = 47 \Omega + j80 \Omega - j35 \Omega = 47 \Omega + j45 \Omega = 65.1\angle 43.8^\circ \Omega}$

$$\mathbf{I_T = \frac{V_s}{Z_T} = \frac{4\angle 0^\circ \text{ V}}{65.1\angle 43.8^\circ \Omega} = 61.4\angle -43.8^\circ \text{ mA}}$$

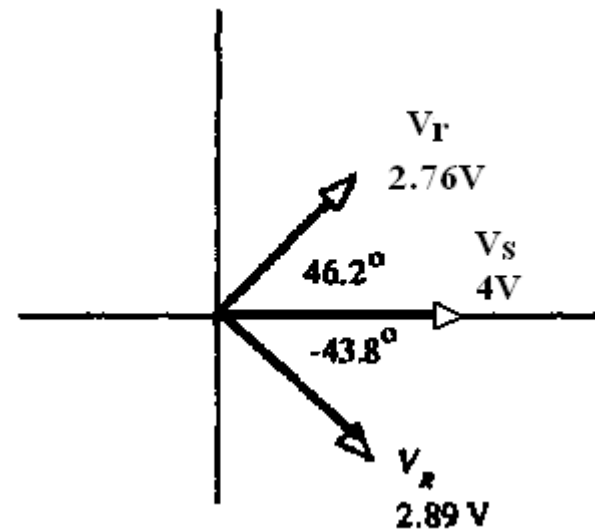
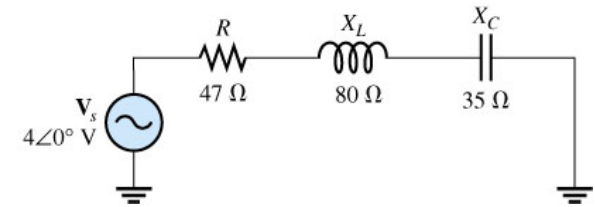
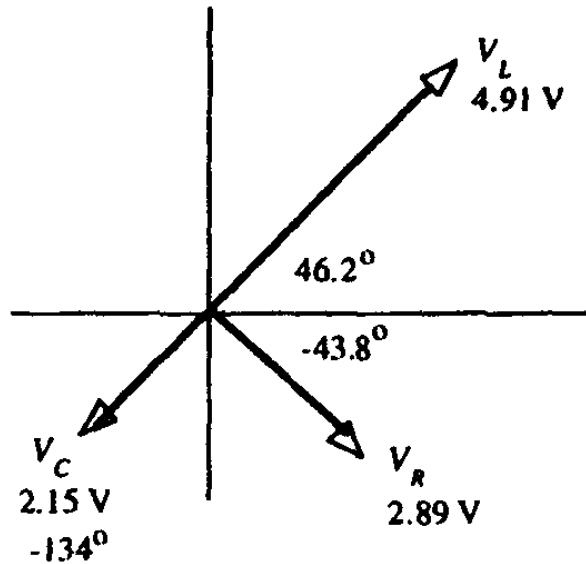
$$\mathbf{V_R = I_T R = (61.4\angle -43.8^\circ \text{ mA})(47\angle 0^\circ \Omega) = 2.89\angle -43.8^\circ \text{ V}}$$

$$\mathbf{V_L = I_T X_L = (61.4\angle -43.8^\circ \text{ mA})(80\angle 90^\circ \Omega) = 4.91\angle 46.2^\circ \text{ V}}$$

$$\mathbf{V_C = I_T X_C = (61.4\angle -43.8^\circ \text{ mA})(35\angle -90^\circ \Omega) = 2.15\angle -134^\circ \text{ V}}$$

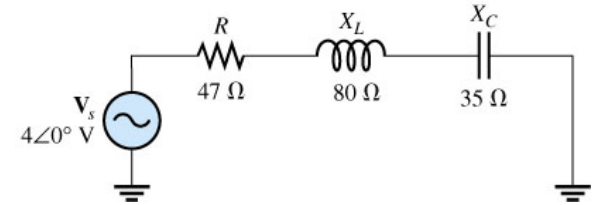
Series RLC Example 2 (cont.)

b).



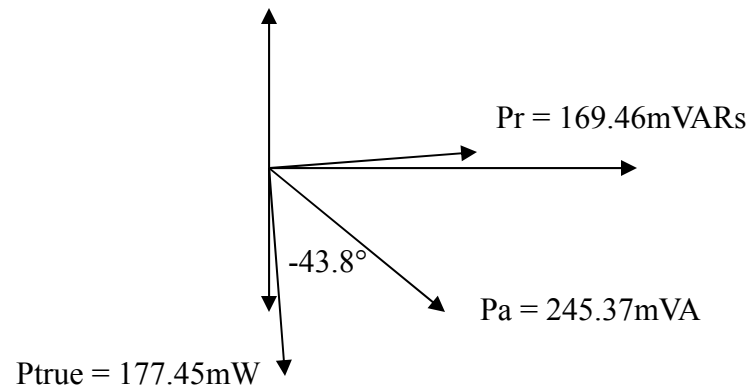
- **Note:** $V_r = V_{CL} = V_{\text{tot}} = (V_L - V_C)$

Series RLC Example 2 (cont.)



c).

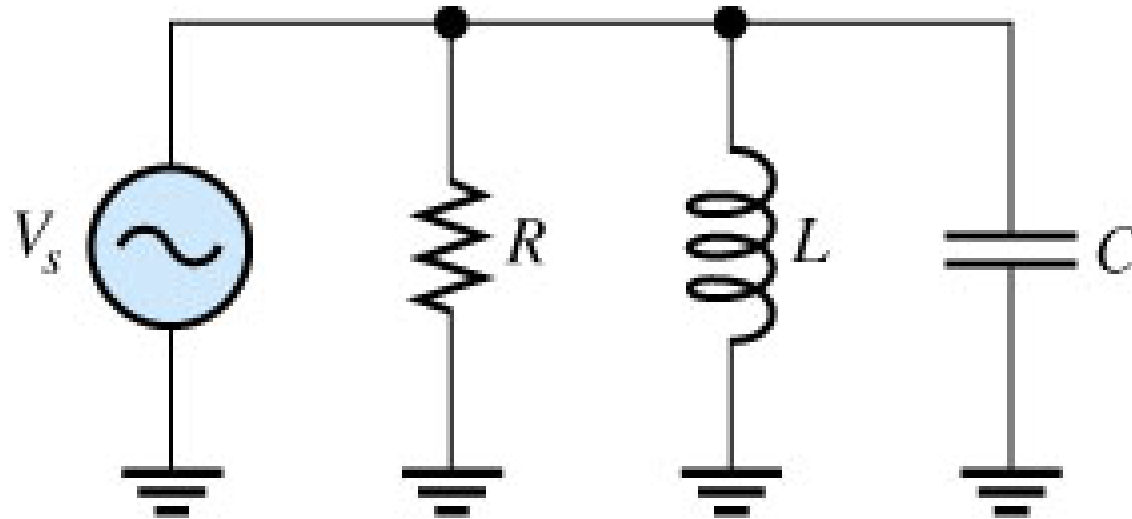
- $P_{\text{true}} = V_R I_T = (61.4\text{mA} \angle -43.8^\circ)(2.89\text{V} \angle -43.8^\circ) = \underline{177.45\text{mW} \angle -87.6^\circ}$
- $P_r = V_r I_T = (V_L - V_C)(I_T) = [(4.91\text{V} - 2.15\text{V}) \angle 46.2^\circ](61.4\text{mA} \angle -43.8^\circ)$
 $= (2.76\text{V} \angle 46.2^\circ)(61.4\text{mA} \angle -43.8^\circ) = \underline{169.46\text{mVARs} \angle 2.4^\circ}$
- $P_a = \sqrt{P_{\text{true}}^2 + P_r^2} = \sqrt{(177.45\text{mW})^2 + (169.46\text{mVARs})^2}$
 $= \underline{245.37\text{mVA} \angle -43.8^\circ}$
- $P_F = \text{real-power/apparent power} = (177.45\text{mW})/(245.37\text{mVA}) = \underline{0.723 \text{ lagging}}$



Impedance of Parallel RLC Circuits

The total impedance of a parallel RLC circuit is:

$$1/\mathbf{Z} = 1/R\angle 0^\circ + 1/X_L\angle 90^\circ + 1/X_C\angle -90^\circ$$



Conductance, Susceptance, and Admittance

- Conductance:

$$\mathbf{G} = 1/R \angle 0^\circ = G \angle 0^\circ$$

- Capacitive Susceptance:

$$\mathbf{B}_C = 1/(X_C \angle -90^\circ) = B_C \angle 90^\circ = jB_C$$

- Inductive Susceptance:

$$\mathbf{B}_L = 1/(X_L \angle 90^\circ) = B_L \angle -90^\circ = -jB_L$$

- Admittance:

$$\mathbf{Y} = 1/(Z \angle \pm\theta) = Y \angle \pm\theta = G + jB_C - jB_L$$

Analysis of Parallel RLC Circuits

- The smaller reactance in a parallel circuit dominates because it results in the larger branch current.
- At low frequencies $X_L < X_C$; therefore the circuit is inductive.
- **Parallel resonance** is the point where $X_L = X_C$
- At high frequencies $X_C < X_L$; therefore the circuit is capacitive.

Current Relationships

- Total Current can be expressed as:

$$I_{\text{tot}} = \sqrt{I_R^2 + (I_C - I_L)^2} \angle \tan^{-1}(I_{CL}/I_R)$$

- Typical current phasor diagram for a parallel RLC

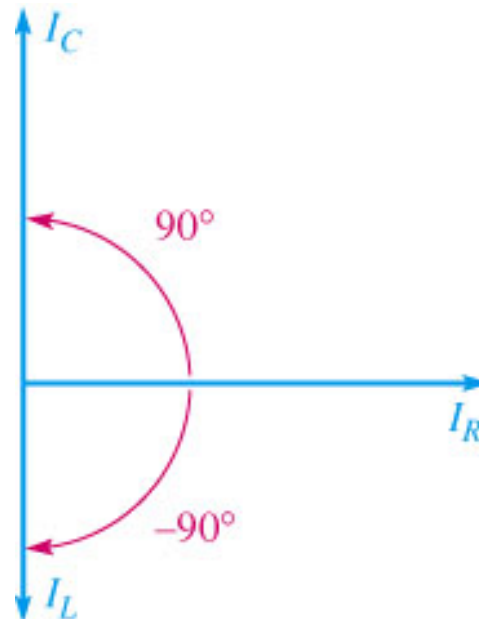
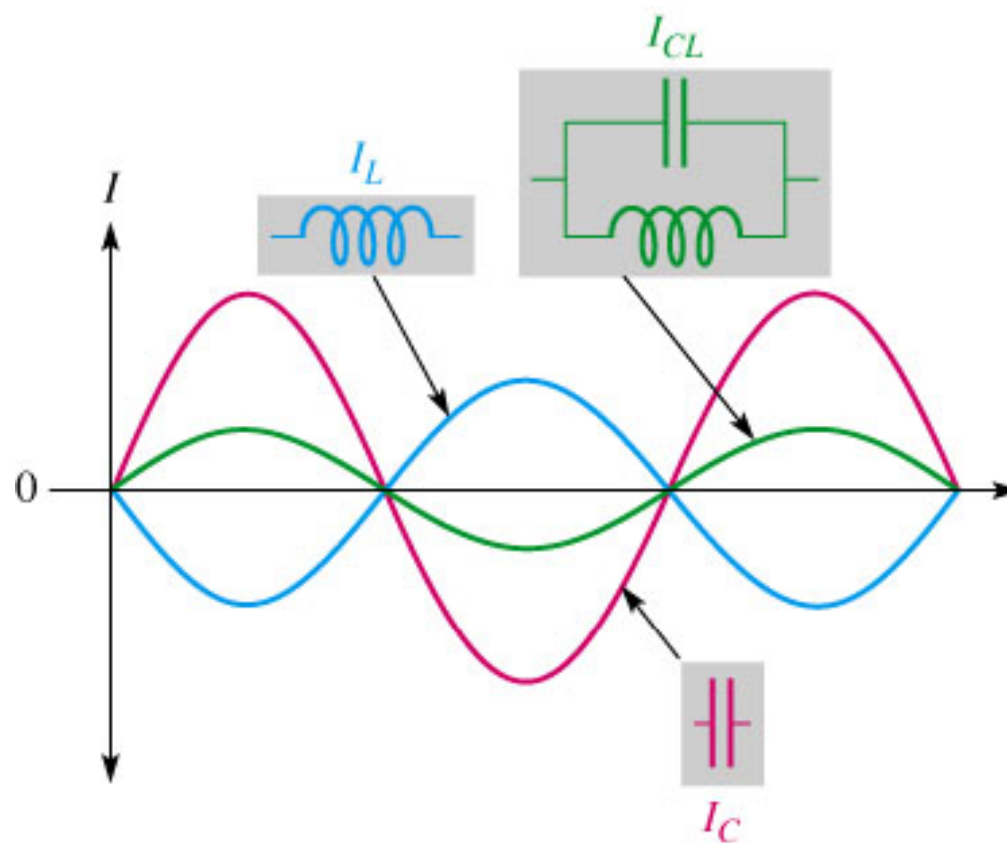


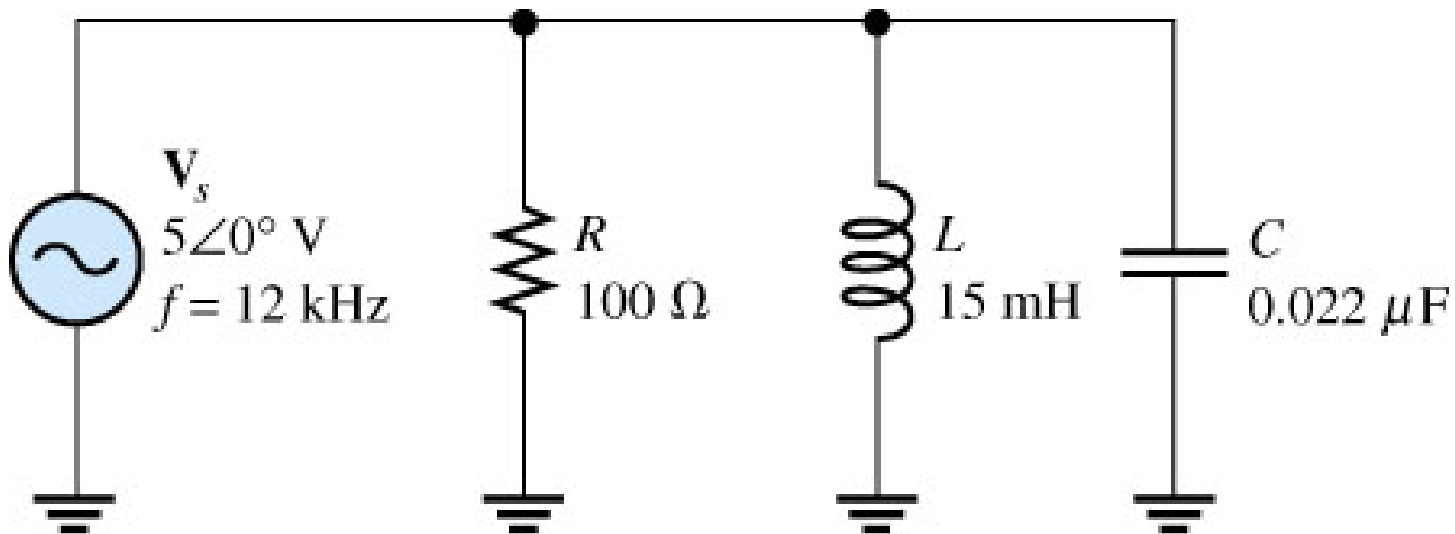
Figure 18--23 I_C and I_L effectively subtract.



PARALLEL RLC EXAMPLE 1

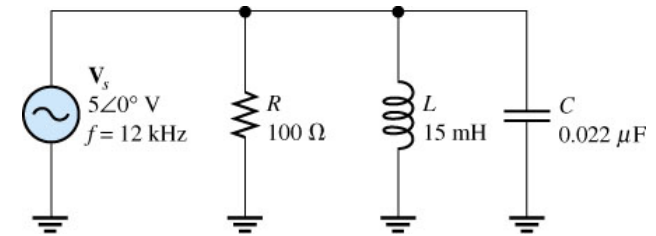
Find the following parameters:

- Circuit Impedance in polar form.
- Is the circuit more inductive or more capacitive?
- At what frequency does the circuit change its reactive characteristic from inductive to capacitive or vice-versa?



PARALLEL RLC EXAMPLE 1 (cont.)

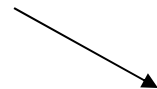
a). Circuit Impedance in polar form.



$$X_L = 2\pi fL = 2\pi(12 \text{ kHz})(15 \text{ mH}) = 1131 \Omega$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(12 \text{ kHz})(0.022 \mu\text{F})} = 603 \Omega$$

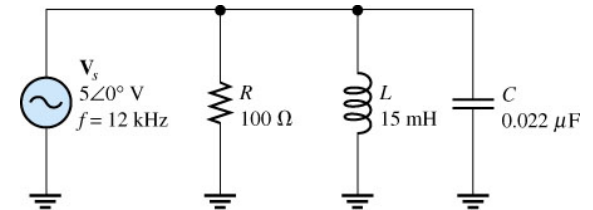
$$\begin{aligned} \mathbf{Z} &= \frac{1}{\frac{1}{100\angle 0^\circ \Omega} + \frac{1}{1131\angle 90^\circ \Omega} + \frac{1}{603\angle -90^\circ \Omega}} \\ &= \frac{1}{10 \text{ mS} - j0.884 \text{ mS} + j1.66 \text{ mS}} = 99.7\angle -4.43^\circ \Omega \end{aligned}$$



$$\mathbf{Y}_T = 10 \text{ mS} + j0.776 \text{ mS} = 0.01003 \text{ S} \angle 4.43^\circ$$

$$\mathbf{Z}_T = 1/\mathbf{Y}_T = 1/(0.01003 \text{ S} \angle 4.43^\circ) = 99.7 \text{ ohms} \angle -4.43^\circ$$

PARALLEL RLC EXAMPLE 1 (cont.)



b). Since θ of $Z_T = -4.43^\circ$, the circuit is slightly capacitive.

c).

$$X_L < X_C$$

$$2\pi fL < \frac{1}{2\pi fC}$$

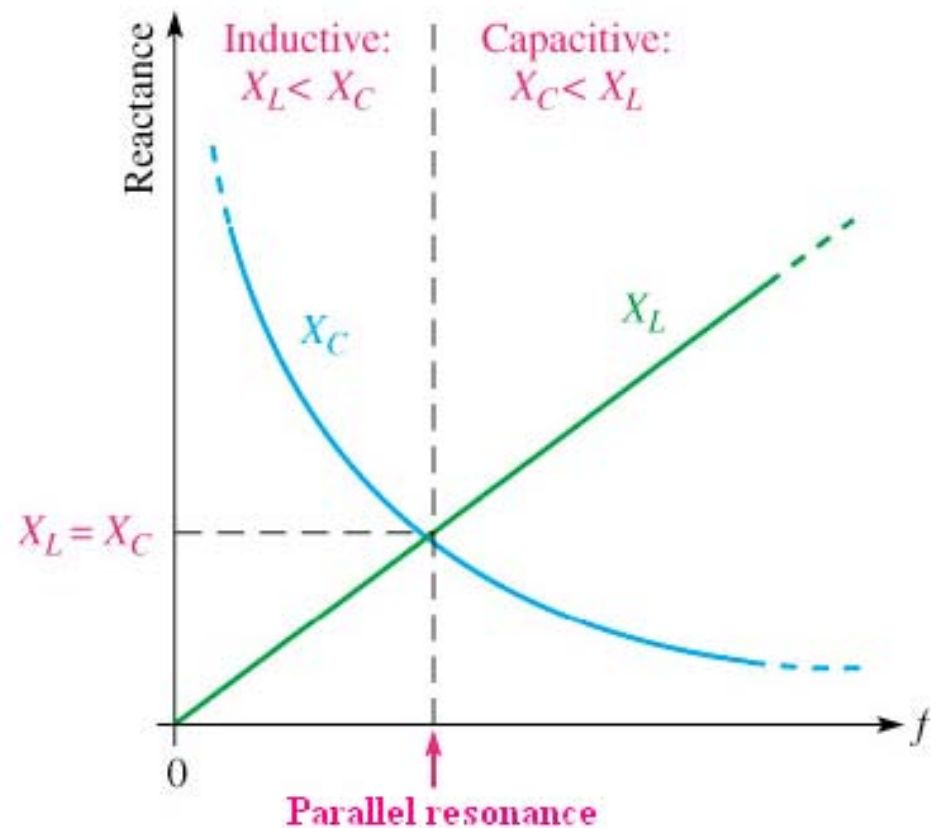
$$f^2 < \frac{1}{4\pi^2 LC}$$

$$f < \frac{1}{\sqrt{4\pi^2 LC}}$$

$$f < \frac{1}{2\pi\sqrt{LC}}$$

$$f < \frac{1}{2\pi\sqrt{(15 \text{ mH})(0.022 \mu\text{F})}}$$

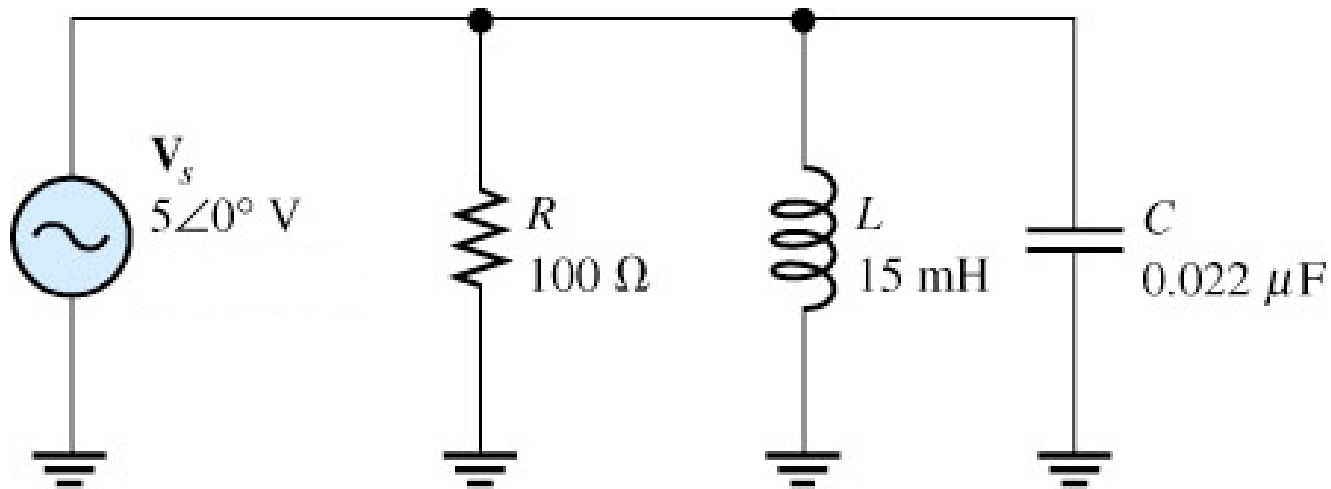
$$f < 8.76 \text{ kHz}$$



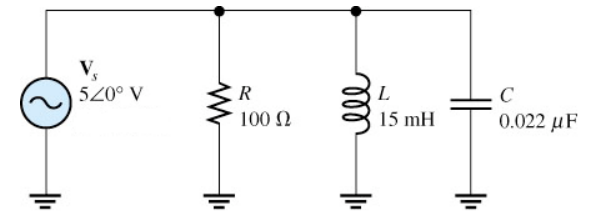
PARALLEL RLC EXAMPLE 2

Find the following parameters:

- All currents and voltages in polar form if the source frequency is 12KHz.
- The current phasor diagram.



PARALLEL RLC EXAMPLE 2 (cont.)



a). All currents and voltages in polar form.

$$\mathbf{I_T} = \frac{\mathbf{V_s}}{\mathbf{Z}} = \frac{5\angle 0^\circ \text{ V}}{99.7\angle -4.43^\circ \Omega} = \mathbf{50.2\angle 4.43^\circ \text{ mA}}$$

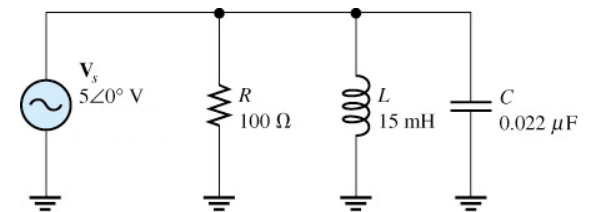
$$\mathbf{I_R} = \frac{\mathbf{V_s}}{\mathbf{R}} = \frac{5\angle 0^\circ \text{ V}}{100\angle 0^\circ \Omega} = \mathbf{50\angle 0^\circ \text{ mA}}$$

$$\mathbf{I_L} = \frac{\mathbf{V_s}}{\mathbf{X_L}} = \frac{5\angle 0^\circ \text{ V}}{1131\angle 90^\circ \Omega} = \mathbf{4.42\angle -90^\circ \text{ mA}}$$

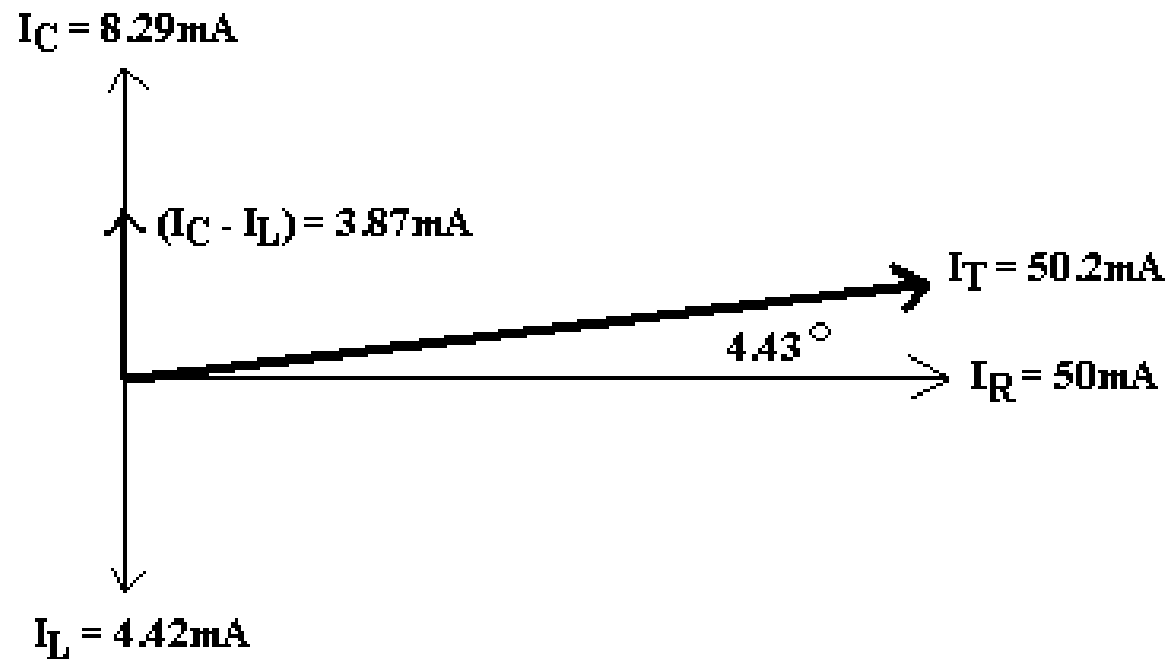
$$\mathbf{I_C} = \frac{\mathbf{V_s}}{\mathbf{X_C}} = \frac{5\angle 0^\circ \text{ V}}{603\angle -90^\circ \Omega} = \mathbf{8.29\angle 90^\circ \text{ mA}}$$

$$\mathbf{V_R} = \mathbf{V_L} = \mathbf{V_C} = \mathbf{5\angle 0^\circ \text{ V}}$$

PARALLEL RLC EXAMPLE 2 (cont.)



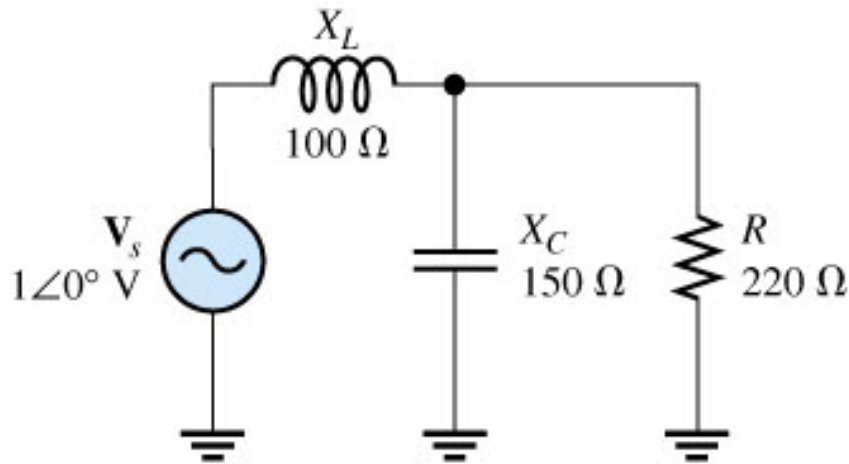
b). Current phasor diagram:



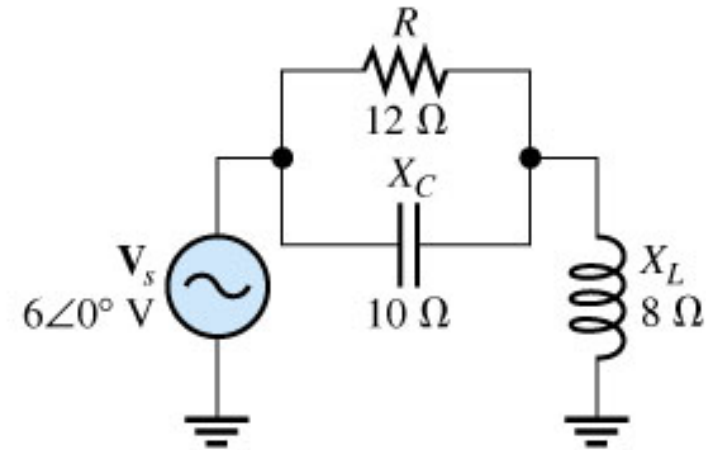
SERIES-PARALLEL RLC

EXAMPLE 1

- Find the total impedance for each circuit below:



(a)

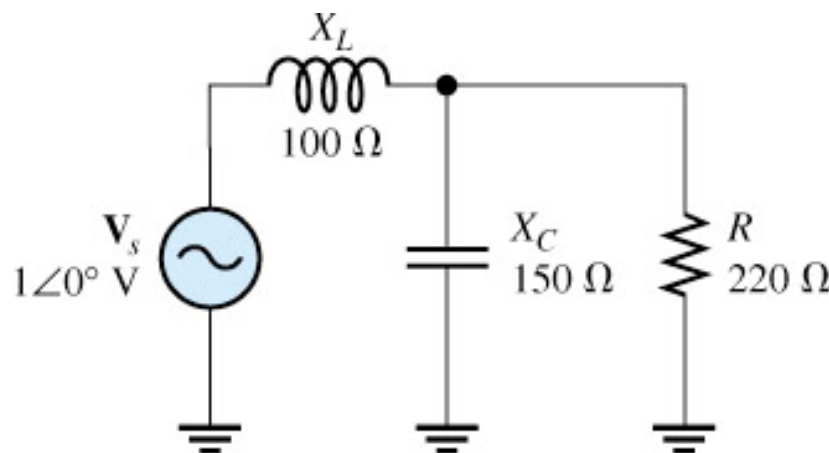


(b)

SERIES-PARALLEL RLC

EXAMPLE 1 (cont.)

(a)

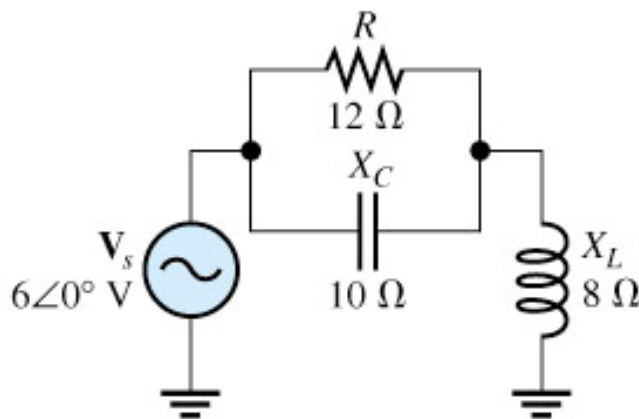


$$\begin{aligned} \mathbf{Z}_T &= jX_L + \frac{R(-jX_C)}{R - jX_C} \\ &= j100\ \Omega + \frac{(220\ \Omega)(-j150\ \Omega)}{220\ \Omega - j150\ \Omega} = j100\ \Omega + \frac{-j33000\ \Omega}{266\angle -34.3^\circ} = j100\ \Omega + 124\angle -55.7^\circ\ \Omega \\ &= j100\ \Omega + 69.9\ \Omega - j102.4\ \Omega = 69.9\ \Omega - j2.4\ \Omega = 69.9\angle -1.97^\circ\ \Omega \end{aligned}$$

SERIES-PARALLEL RLC

EXAMPLE 1 (cont.)

(b)

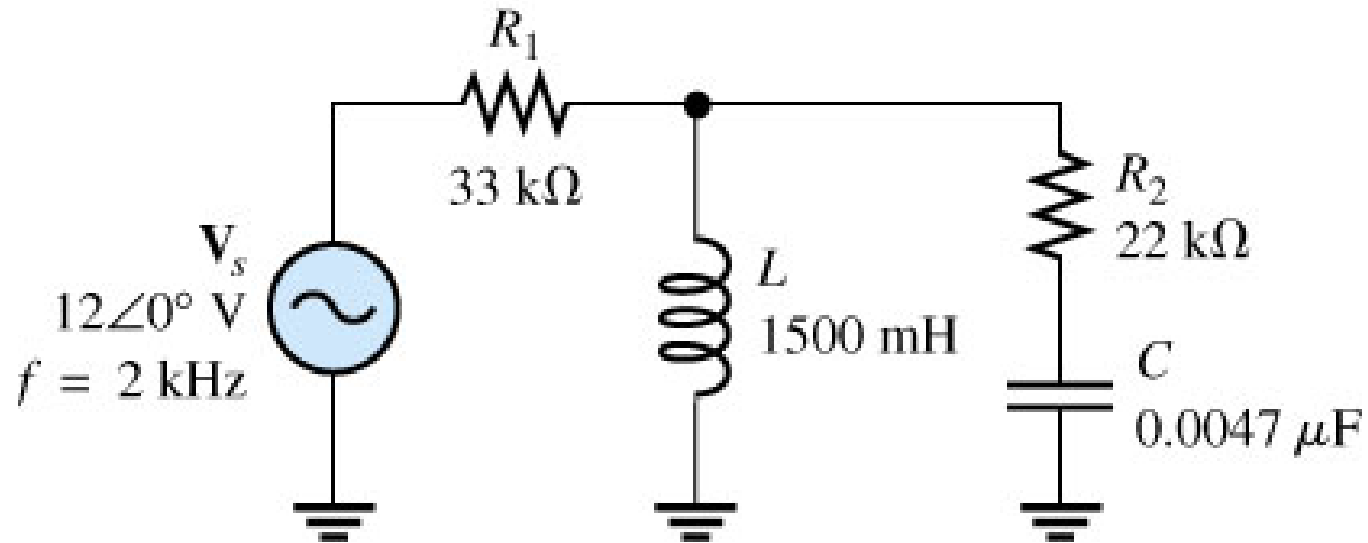


$$\begin{aligned} \mathbf{Z_T} &= jX_L + \frac{R(-jX_C)}{R - jX_C} \\ &= j8\ \Omega + \frac{-j120\ \Omega}{15.6\angle -39.8^\circ} = j8\ \Omega + 7.69\angle -50.2^\circ\ \Omega \\ &= j8\ \Omega + 4.92\ \Omega - j5.91\ \Omega = 4.92\ \Omega + j2.09\ \Omega = \mathbf{5.35\angle 23.0^\circ\ \Omega} \end{aligned}$$

SERIES-PARALLEL RLC

EXAMPLE 2

- For the circuit below, find the following parameters:
 - (a) The voltage across each element in polar form.
 - (b) Convert the circuit to an equivalent series form.



SERIES-PARALLEL RLC EXAMPLE 2 (cont.)

(a) The voltage across each element in polar form.

$$X_L = 2\pi(2 \text{ kHz})(1500 \text{ mH}) = 18.9 \text{ k}\Omega$$

$$X_C = \frac{1}{2\pi(2 \text{ kHz})(0.0047 \text{ }\mu\text{F})} = 16.9 \text{ k}\Omega$$

$$\begin{aligned} \mathbf{Z}_{\text{LR2C}} &= jX_L \parallel (R_2 - jX_C) = \frac{jX_L(R_2 - jX_C)}{R_2 - jX_C + jX_L} \\ &= \frac{(18.9 \angle 90^\circ \text{ k}\Omega)(27.7 \angle -37.5^\circ \text{ k}\Omega)}{22.1 \angle 5.19^\circ \text{ k}\Omega} = 23.7 \angle 47.3^\circ \text{ k}\Omega = 16.1 \text{ k}\Omega + j17.4 \text{ k}\Omega \end{aligned}$$

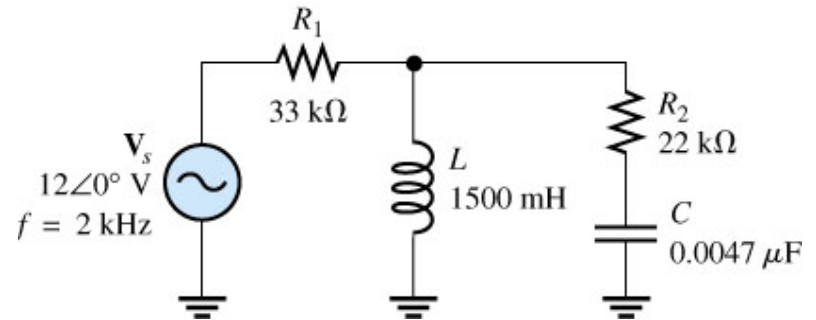
$$\mathbf{Z}_T = \mathbf{R}_1 + \mathbf{Z}_{\text{LR2C}} = 33 \text{ k}\Omega + 16.1 \text{ k}\Omega + j17.4 \text{ k}\Omega = 49.1 \text{ k}\Omega + j17.4 \text{ k}\Omega = 52.1 \angle 19.5^\circ \text{ k}\Omega$$

$$\mathbf{V}_{R1} = \left(\frac{\mathbf{R}_1}{\mathbf{Z}_T} \right) \mathbf{V}_s = \left(\frac{33 \angle 0^\circ \text{ k}\Omega}{52.1 \angle 19.5^\circ \text{ k}\Omega} \right) 12 \angle 0^\circ \text{ V} = 7.60 \angle -19.5^\circ \text{ V}$$

$$\mathbf{V}_L = \left(\frac{\mathbf{Z}_{\text{LR2C}}}{\mathbf{Z}_T} \right) \mathbf{V}_s = \left(\frac{23.7 \angle 47.3^\circ \text{ k}\Omega}{52.1 \angle 19.5^\circ \text{ k}\Omega} \right) 12 \angle 0^\circ \text{ V} = 5.46 \angle 27.8^\circ \text{ V}$$

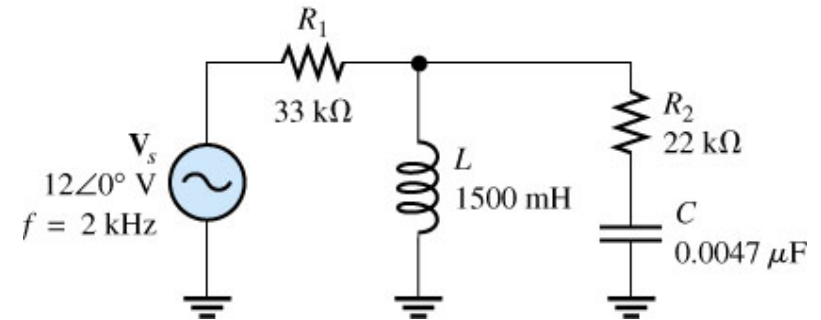
$$\mathbf{V}_{R2} = \left(\frac{\mathbf{R}_2}{R_2 - jX_C} \right) \mathbf{V}_L = \left(\frac{22 \angle 0^\circ \text{ k}\Omega}{27.7 \angle -37.5^\circ \text{ k}\Omega} \right) 5.46 \angle 27.8^\circ \text{ V} = 4.34 \angle 63.3^\circ \text{ V}$$

$$\mathbf{V}_C = \left(\frac{\mathbf{X}_C}{R_2 - jX_C} \right) \mathbf{V}_L = \left(\frac{16.9 \angle -90^\circ \text{ k}\Omega}{27.7 \angle -37.5^\circ \text{ k}\Omega} \right) 5.46 \angle 27.8^\circ \text{ V} = 3.33 \angle -24.7^\circ \text{ V}$$



SERIES-PARALLEL RLC EXAMPLE 2 (cont.)

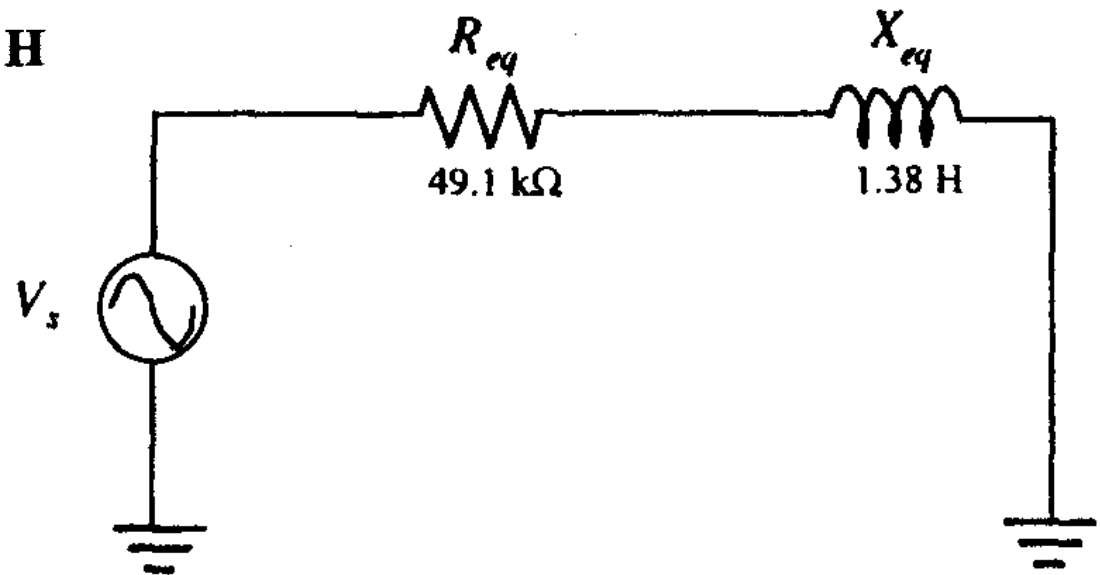
(b) Equivalent series form:



$$\mathbf{Z_T} = \mathbf{R_1} + \mathbf{Z_2} = 33 \text{ k}\Omega + 16.1 \text{ k}\Omega + j17.4 \text{ k}\Omega = 49.1 \text{ k}\Omega + j17.4 \text{ k}\Omega = R_{eq} + jX_{eq}$$

$$X_{eq} = 2\pi fL$$

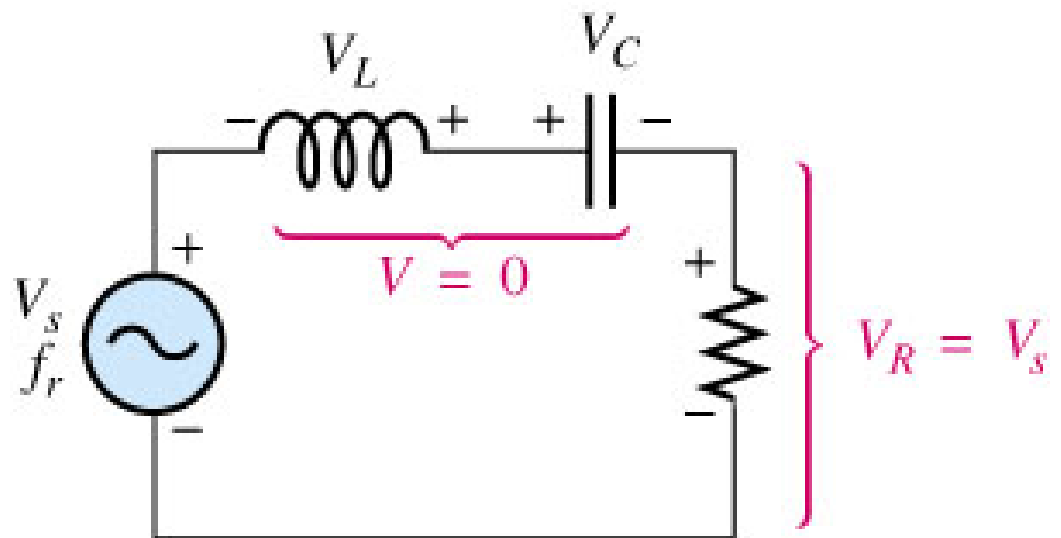
$$L = \frac{X_{eq}}{2\pi f} = \frac{17.4 \text{ k}\Omega}{2\pi(2 \text{ kHz})} = 1.38 \text{ H}$$



Series Resonance

- Resonance is a condition in a series RLC circuit in which the capacitive and inductive reactances are equal in magnitude.
- The result is a purely resistive impedance.
- The formula for series resonance is:

$$f_r = 1/(2\pi\sqrt{LC})$$



Current and Voltage in a Series RLC Circuit

- At the series resonant frequency, the current is maximum ($I_{\max} = V_s/R$).
- Above and below resonance, the current decreases because the impedance increases.
- Resistor voltage is maximum at resonance.
- The voltages across L and C are maximum at resonance, but they are also equal in magnitude and 180° out of phase, so they cancel (the total voltage across L and C is zero).

Figure 18--13 Series *RLC* impedance as a function of frequency.

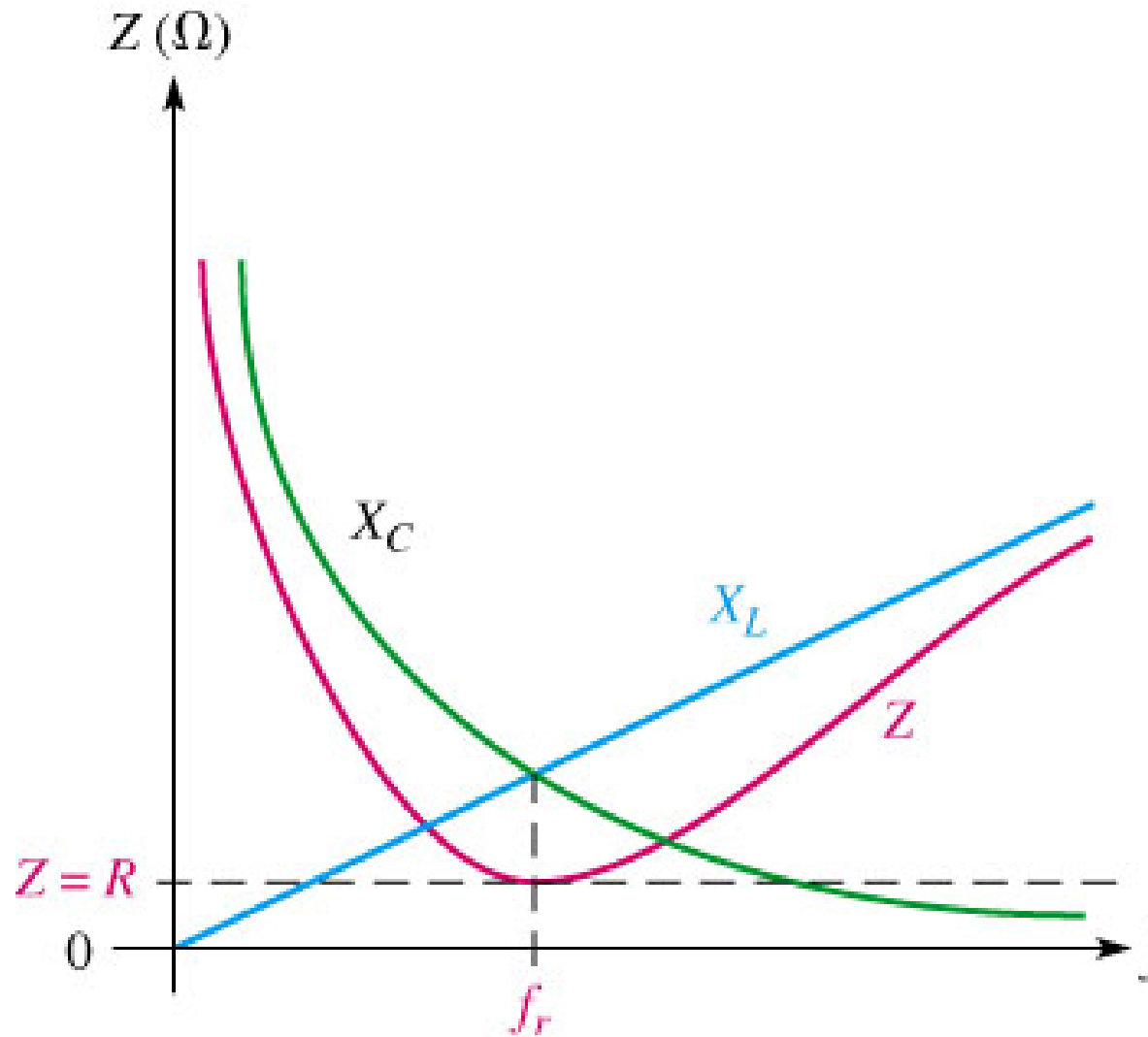
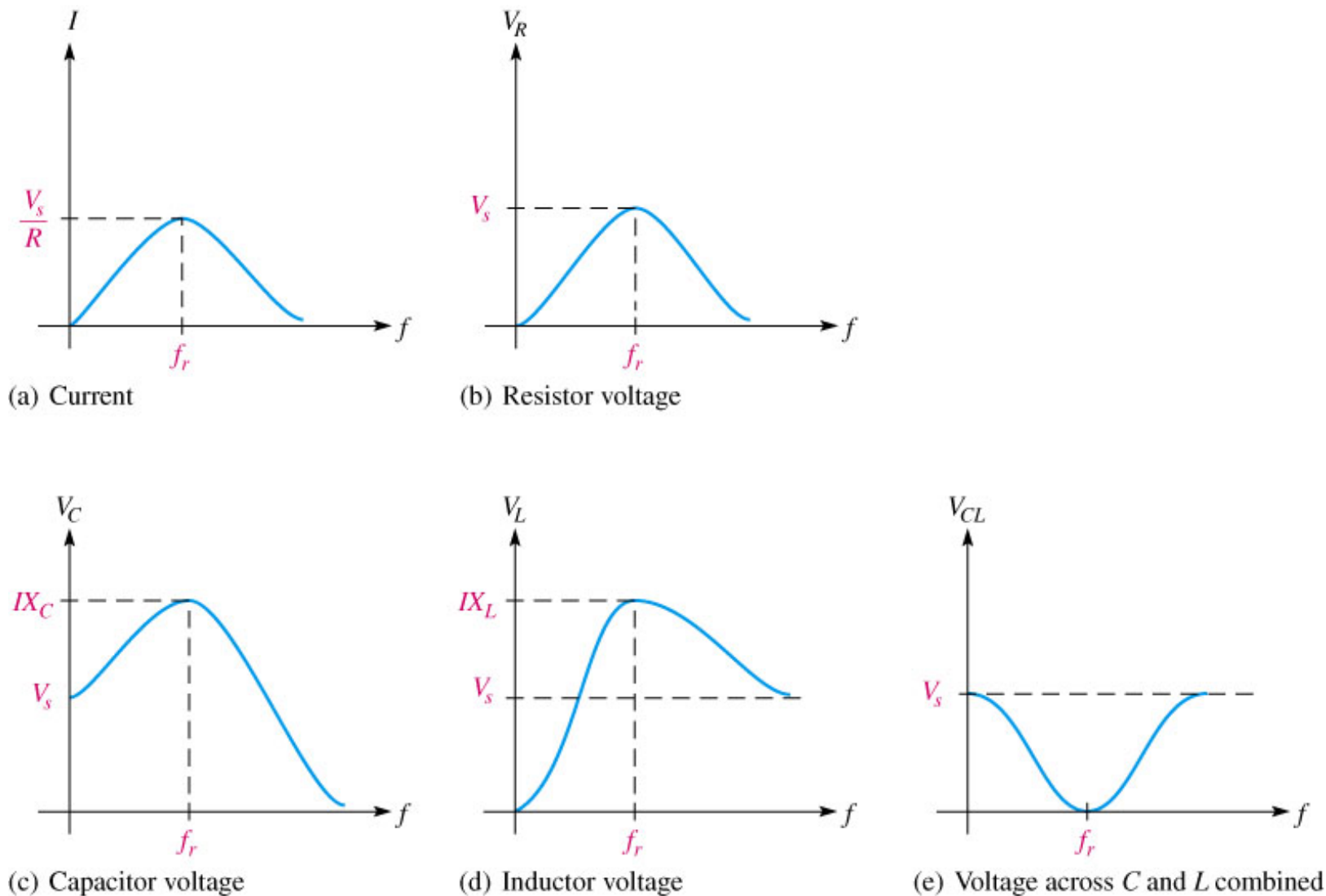
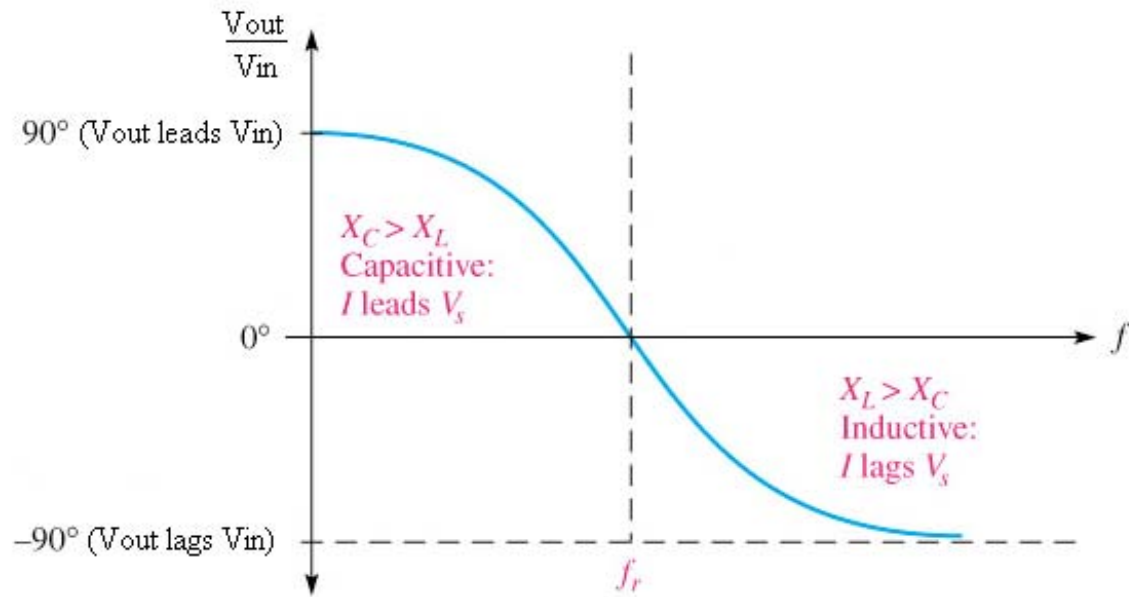
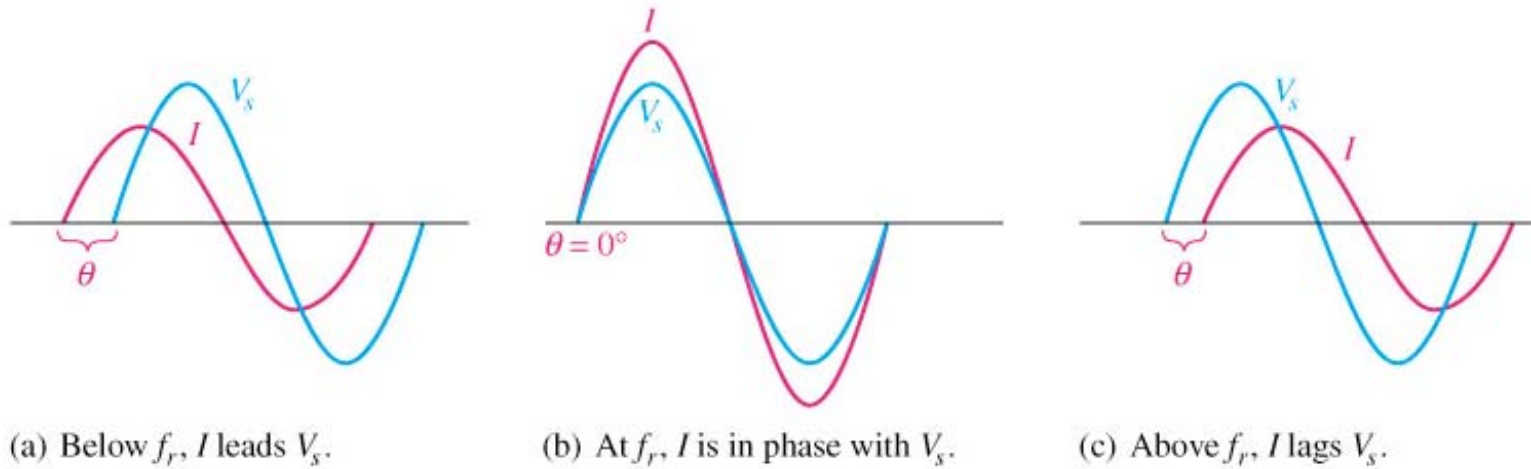


Figure 18–15 Generalized current and voltage magnitudes as a function of frequency in a series RLC circuit. V_C and V_L can be much larger than the source voltage. The shapes of the graphs depend on particular circuit values.

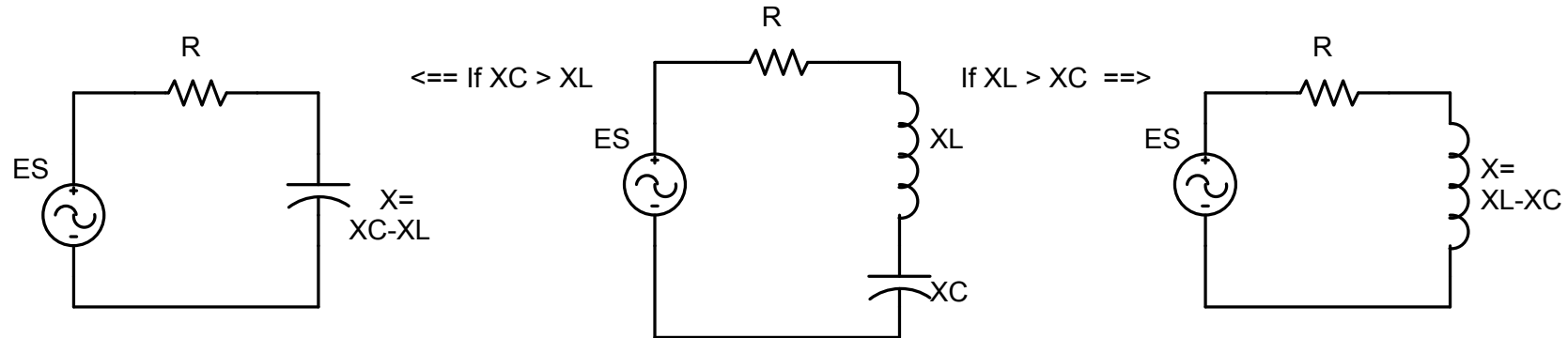


Phase Angle of a Series RLC Circuit



(d) Phase angle versus frequency.

Series Resonant Circuits



- The frequency at which $X_L = X_C$ is also called the series resonant frequency, f_R , and such a circuit is called a series resonant circuit.
- To derive f_R , $X_L = 2\pi f_R L$, $X_C = 1/(2\pi f_R C)$, substitute in $X_L = X_C$ and solve for f_R .

$$f_R = 1/(2\pi\sqrt{LC})$$
- Note several concepts:
 1. There will be a resonant frequency for each value of L and C where $X_L = X_C$
 2. f_R does NOT depend on the value of circuit resistance.

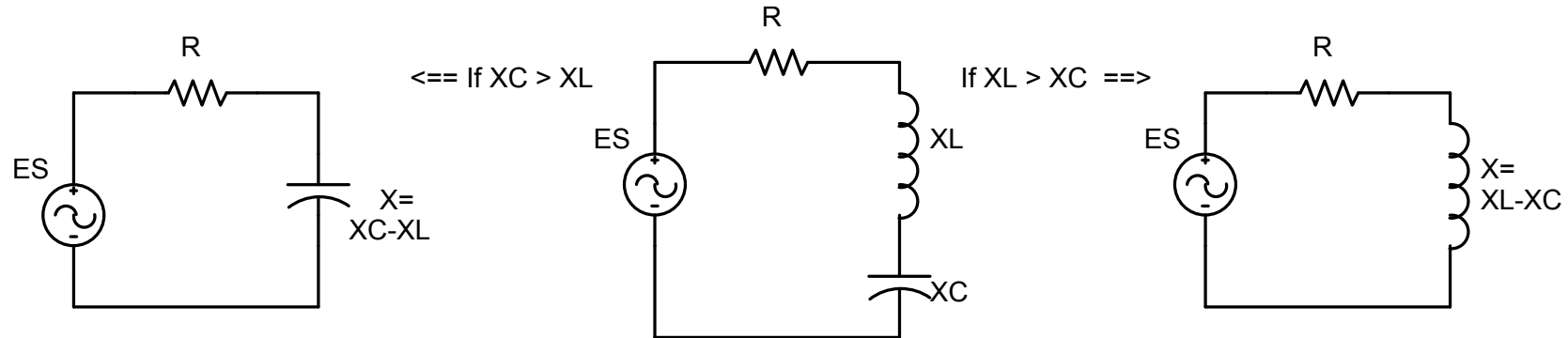
Series Resonant Circuits (cont.)

- Now, just what is this resonant circuit? What does it do? What is it used for?
 - a. **First, it is intended to operate at or near f_R .**
 - b. **At f_R , $P_F = 1$**
 - c. **At f_R , phase shift = 0°**
 - d. **When operating, AC is exchanged back and forth between capacitor and inductor most efficiently at f_R , like an echo. This is what is meant by resonant or resonates = reproduce or echo a signal.**

Series Resonant circuits are used for:

1. **Tuned circuits in radio and TV receivers to select or reject a desired or undesired frequency or narrow range of frequencies. So they would be used to select a desired frequency from the multitude of frequencies at the antenna. A subcategory of tuned circuits is pre-select circuits (similar to tuned circuits).**
2. **Traps: Output stages of transmitters produce desired frequency plus a lower power but undesired 2nd harmonic (2X transmitter frequency). A series resonant R(LC) circuit can select and short (shunt) the undesired 2nd harmonic to ground before it goes out the transmitter antenna. (Note that these applications are actually special types of “filters” which are circuits that are designed to pass certain frequencies while rejecting or blocking other frequencies.**

Series Resonant Circuits (cont.)



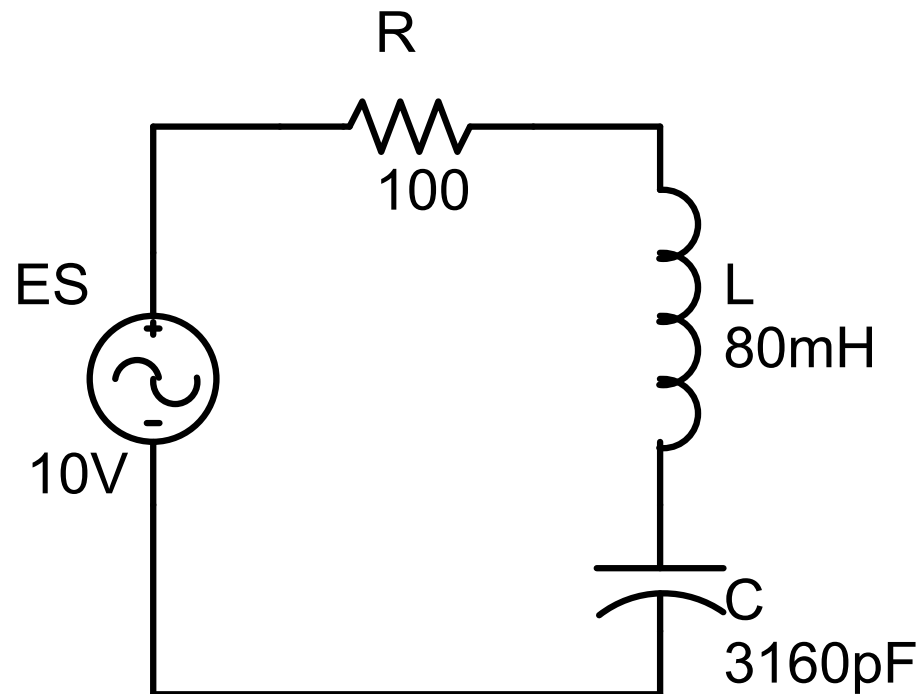
- Now, looking at the series resonant circuit, let's identify and examine the important characteristics that allow it to provide the applications described above.

1. First, let's summarize what happens to Z above, below, and at the resonant frequency:

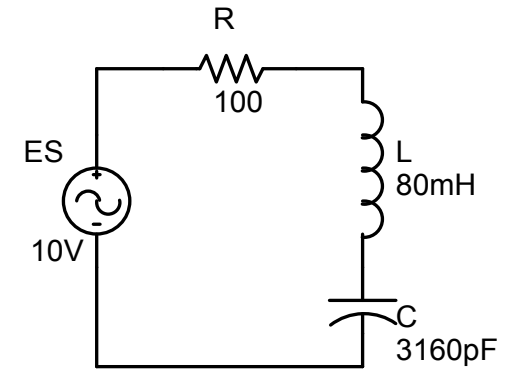
- as f goes below f_R ($f < f_R$) $X_L \downarrow$ and $X_C \uparrow$ so the circuit is more capacitive
 - as f goes above f_R ($f > f_R$) $X_L \uparrow$ and $X_C \downarrow$ so the circuit is more inductive
 - when $f = f_R$, $X_L = X_C$ and cancel at 180° so circuit Z is resistive ($Z = R$)
 - when $Z = R$, circuit Z is **MINIMUM** at the resonant frequency
- If Circuit Z is **MIN** at f_R , what is true about I ? I becomes **MAX** at f_R .
 - Since circuit is purely resistive at f_R , then circuit I is in phase ($@0^\circ$) with E at f_R .

Series Resonant Example 1

- Given a series RLC circuit with $E_s = 10\text{V}$, $R = 100\Omega$, $L = 80\text{mH}$, & $C = 3160\text{pF}$.
- Find f_R , X_L , X_C , Z , I , V_R , V_L , V_C at f_R ; at $f = 0.9f_R$; & at $f = 1.1f_R$



Series Resonant Example 1 (cont.)



Solution:

$$f_R = 1/(2\pi\sqrt{LC}) = 10\text{kHz}$$

$$@ f = f_R, X_C = X_L$$

$$X_L = 2\pi f_R L = 5\text{k}\Omega$$

$$\text{Net } X = 0$$

$$I = E_S/Z = 100\text{mA}@0^\circ$$

$$V_R = IR = 10\text{V}$$

$$V_C = IX_C = 500\text{V}$$

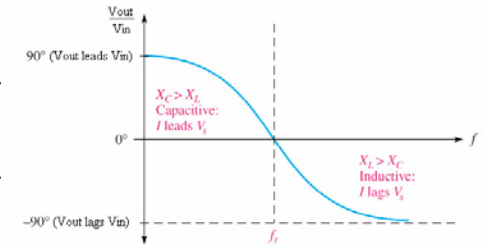
$$X_C = 1/(2\pi f_R C) = 5\text{k}\Omega$$

$$Z = R = 100\Omega$$

$$V_L = IX_L = 500\text{V}$$

- Note how much larger V_L and V_C are than the input voltage, E_S . This is called Voltage Magnification and is designated by Q . So $Q = V_L / E_S$ or $Q = V_C / E_S$

Series Resonant Example 1 (



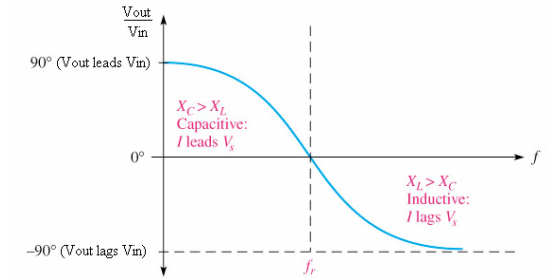
Let's set up a Table/Chart to record and compare values at f_R , $0.9f_R$, and $1.1f_R$:

	Freq.	X_L	X_C	Z	I	$\angle \theta^\circ$ (V_{out}/V_{in})	V_R	V_L	V_C
	$0.9f_R$								
	f_R								
	$1.1f_R$								

@ f_R (resistive)

	Freq.	X _L	X _C	Z	I	∠ θ° (Vout/ Vin)	V _R	V _L	V _C
0.9f _R									
f _R	10kHz	5kΩ	5kΩ	100Ω	100mA	0°	10V	500V	500V
1.1f _R									

Series Resonant Example 1



@ f_R & @ $0.9f_R$ (capacitive)

	Freq.	X_L	X_C	Z	I	$\angle \theta^\circ$ (Vout/Vin)	V_R	V_L	V_C
$0.9f_R$	9kHz	$4.5k\Omega$	$5.5k\Omega$	1005Ω	10mA	$+84^\circ$	1V	45V	55V
f_R	10kHz	$5k\Omega$	$5k\Omega$	100Ω	100mA	0°	10V	500V	500V
$1.1f_R$									

@ f_R & @ $0.9f_R$ & @ $1.1f_R$ (inductive)

	Freq.	X_L	X_C	Z	I	$\angle \theta^\circ$ (Vout/Vin)	V_R	V_L	V_C
$0.9f_R$	9kHz	$4.5k\Omega$	$5.5k\Omega$	1005Ω	10mA	$+84^\circ$	1V	45V	55V
f_R	10kHz	$5k\Omega$	$5k\Omega$	100Ω	100mA	0°	10V	500V	500V
$1.1f_R$	11kHz	$5.5k\Omega$	$4.5k\Omega$	1005Ω	10mA	-84°	1V	55V	45V

Calculating and using Q or Voltage Magnification:

- At resonance, the magnitudes of inductor and capacitor voltages are equal, and much greater than the input source voltage.
 - $Q = V_X/V_{IN}$
 - For the example series RLC circuit previously shown,
 $Q = 500V/10V = \underline{50}$
- Another name for Q, or Voltage Magnification, is Quality Factor (hence the letter “Q”), but can also be called the “selectivity factor”. Generally $Q > 10$ is desirable for good selectivity. There are no units for Q.
- Example 1: $Q = 250$, $V_{IN} = 200\mu V$ @ $f = f_R$. Determine V_C .
 - **Solution:** $V_C = Q \times E_S = \underline{50mV}$
- Since $Q = V_L/E_S = V_C/E_S$, and $V_L = IX_L$, and $V_C = IX_C$, then substitute for V_L and V_C to get:
 - $Q = (IX_L)/E_S = (IX_C)/E_S$
- Also, since $I = E_S/R$ at f_R , then substitute for I and simplify to get $Q = X_L/R = X_C/R$. R in this case, is the total circuit resistance (R_{series} and coil winding resistance R_w). This is referred to as Circuit $Q = Q_{CKT}$.

Q_{coil} and Q_{CKT}

- There are generally two Q's in a tuned circuit.
 - Q_{coil} is X_L/R_W and is the Q of the coil only.
 - Q_{CKT} is X_L/R_T and is the Q of the circuit.
 - The lowest Q dominates and determines a tuned circuit's selectivity and voltage and current magnification ratios.
 - Typically, Q_{CKT} is the lowest because it is the total DC resistance including the coil resistance in the circuit.
 - Example 2: Series RLC circuit, $E_S = 10\text{mV}$, $R_{\text{series}} = 200\Omega$, $R_W = 1 \text{ ohm}$, $L = 40\text{mH}$, $C = 1580\text{pF}$. Find Q_{coil} , Q_{CKT} , and V_L @ resonance.
-
- **Solution:**
 $f_R = 1/(2\pi\sqrt{LC}) = 20\text{kHz}$; $X_L = 2\pi f_R L = 5\text{k}\Omega$; $Q_{\text{coil}} = X_L/R_W = 5000$; $Q_{\text{CKT}} = X_L/R = 5\text{k}/(201\Omega) = 24.88$, $V_L = QE_S = (24.88)(10\text{mV}) = 248.8\text{mV}$

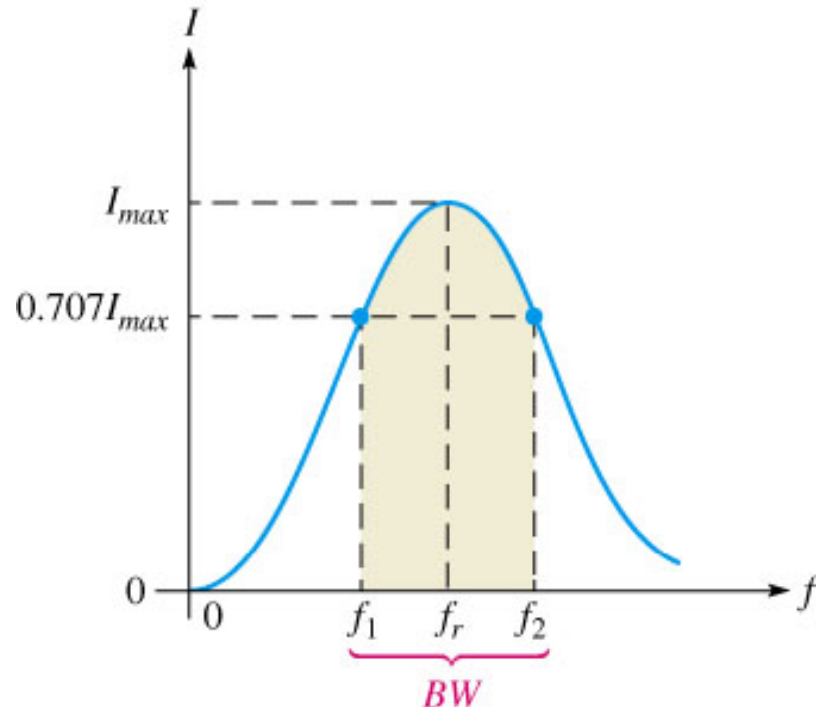
Bandwidth of Series Resonant Circuits

- Current is maximum at resonant frequency.
- Bandwidth (BW) is the range (f_1 to f_2) of frequencies for which the current is greater than 70.7% of the resonant value.

$$BW = f_2 - f_1$$

- Ideally the center frequency is:

$$f_r = (f_1 + f_2)/2$$



- BW for AM Radio: 10KHz, 5KHz on each side of the center frequency.
- BW for FM Radio: 200 KHz, 75 KHz on each side of the center frequency plus an additional 25KHz on each side as a guard band.

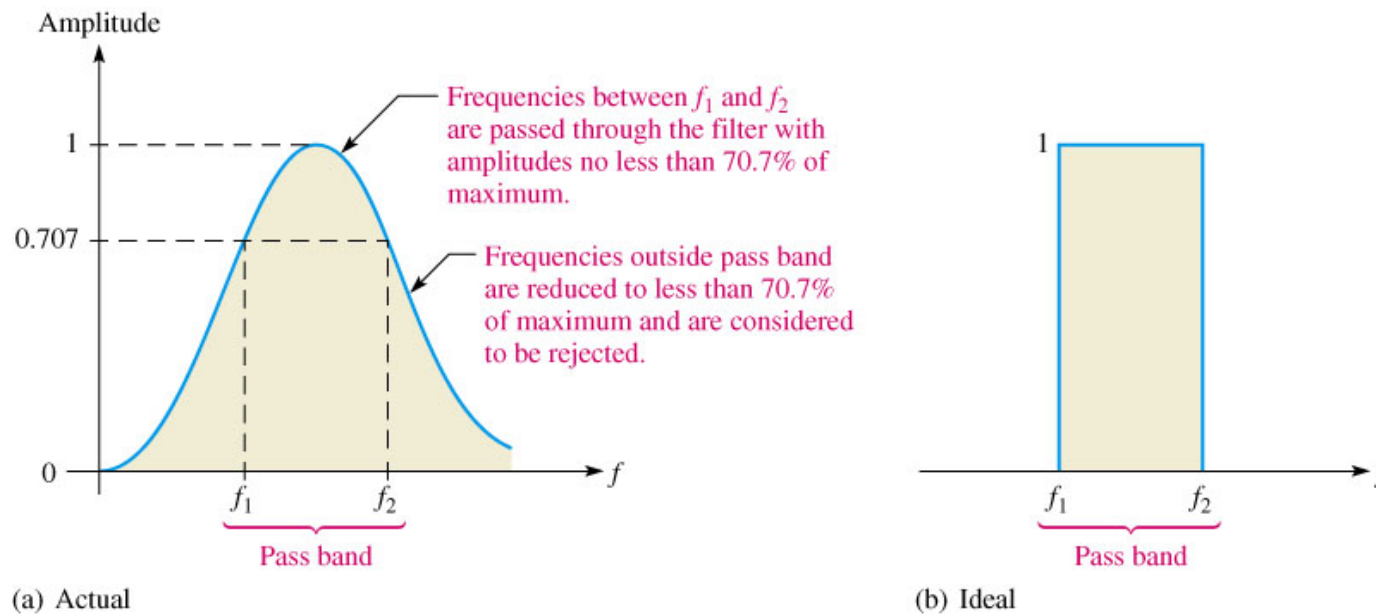
Half-Power Frequencies

- The upper and lower critical frequencies are also called the **half-power frequencies**.
- The power delivered from the source at these frequencies is one-half the power delivered at the resonant frequency.
- $$\begin{aligned} P_{\max} &= I_{\max}^2 R \\ P_{f1} &= P_{f2} = I_{f1}^2 R = (0.707)I_{\max}^2 R \\ &= (0.707)^2 I_{\max}^2 R = 0.5 I_{\max}^2 R = \underline{0.5 P_{\max}} \end{aligned}$$

Selectivity

- Selectivity defines how well a resonant circuit responds to a certain frequency and discriminates against all others.
- The narrower the bandwidth, the greater the selectivity.
- The steeper the slope of the response curve, the greater the selectivity.

Figure 18–46 Generalized selectivity curve of a band-pass filter.

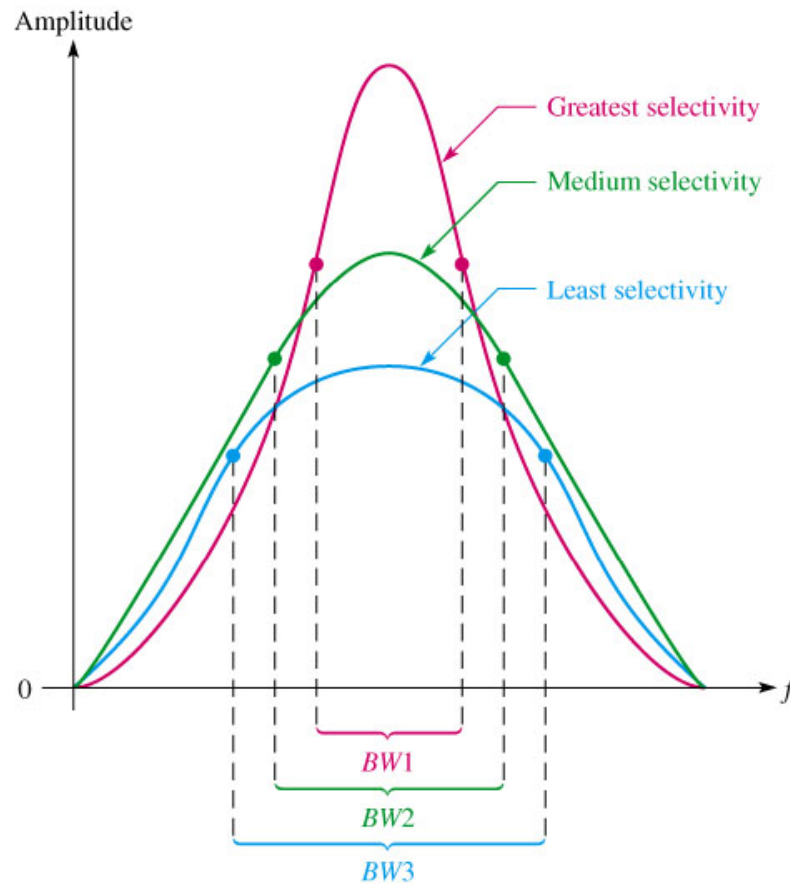


Q Affects Bandwidth

- A higher value of circuit Q results in a narrower bandwidth.
- A lower value of Q causes a wider bandwidth.
- The formula for the bandwidth of a resonant circuit in terms of Q is:

$$BW = f_r/Q$$

Figure 18--47 Comparative selectivity curves.



Series Resonant Example 2

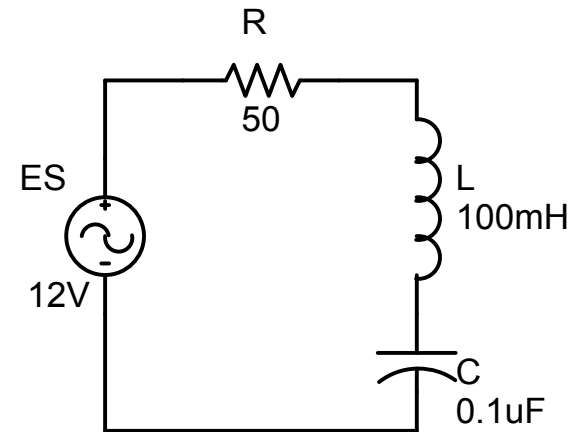
- Find f_R , BW, and Q for the following circuit. Assume R_W and R_{source} are 0 ohms.

Solution:

$$f_R = 1/(2\pi\sqrt{LC}) = 1.6 \text{ kHz}$$

$$Q = X_L/R = 20$$

$$BW = f_R/Q = 80 \text{ Hz}$$



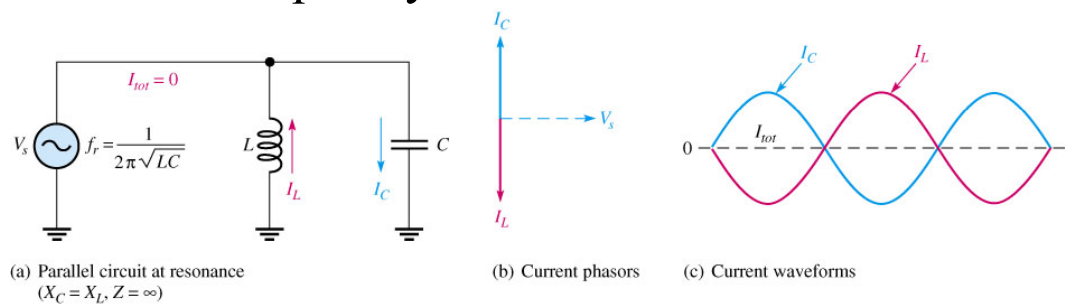
To keep f_R @ 1.6kHz, but double BW to 160Hz, Q becomes 10, so X_L will stay the same but double R to 100Ω.

- Q of a coil = X_L/R_W . There is a practical limit to a coil's Q. If coil $R_W \gg$ circuit R, then Q of coil is essentially Q of circuit. But if circuit R $\gg$ R_W , then Q of circuit is less than Q of coil.

Ideal Parallel Resonance

- Parallel resonance occurs when $X_C = X_L$.
- At resonance, the two branch currents are equal in magnitude, and 180° out-of-phase with each other.
- I_C and I_L cancel phasor-wise. Each component still contains current. Both L and C exchange or circulate current between each other drawing no current from the source.
- Since the total current drawn from the source is zero, the impedance of the ideal parallel LC circuit is infinitely large (∞).
- Ideal (no R_W of coil) parallel resonant frequency:

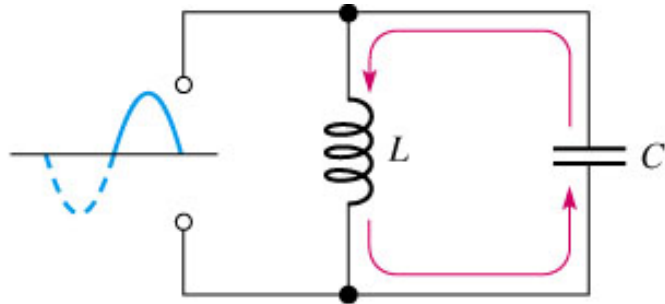
$$f_r = 1/(2\pi\sqrt{LC})$$



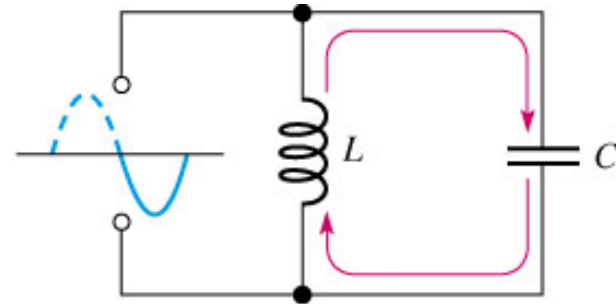
- When $X_C < X_L$ and $I_C > I_L$ circuit is net capacitive.
- When $X_C > X_L$ and $I_C < I_L$ circuit is net inductive.
- Total Impedance at resonance? Since $Z_T = E_S/I_T$, $Z_T = \infty$ ideally at resonance.

Tank Circuit

A parallel resonant circuit stores energy in the magnetic field of the coil and the electric field of the capacitor. The energy is transferred back and forth between the coil and capacitor.

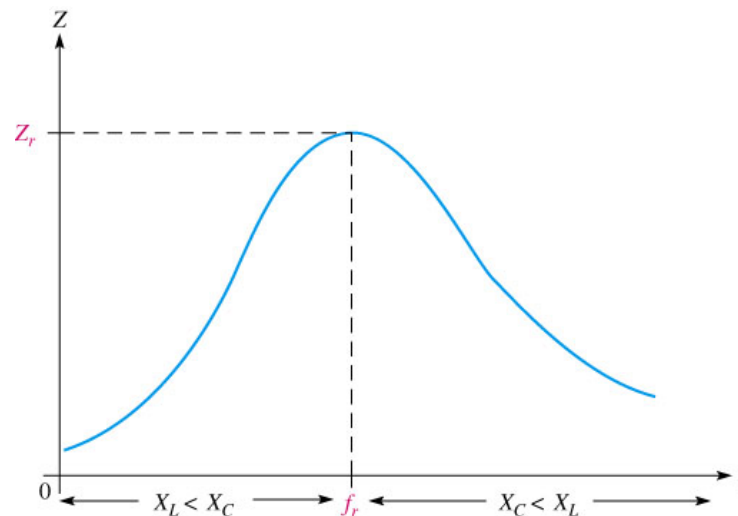


(a) The coil deenergizes as the capacitor charges.



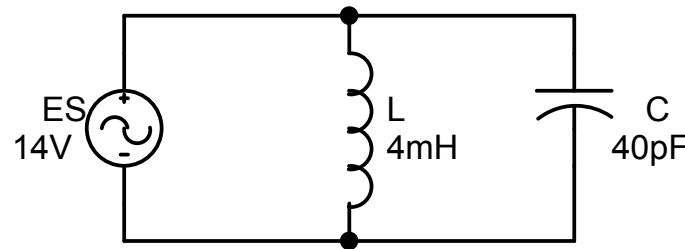
(b) The capacitor discharges as the coil energizes.

Generalized impedance curve for a parallel resonant circuit. The circuit is inductive below f_r , resistive at f_r , and capacitive above f_r .



Ideal Parallel Resonant Circuit Example 1

- Find f_R , X_L , X_C , and I_C , I_L , & I_T at f_R :



Solution:

$$f_R = 1/(2\pi\sqrt{LC}) = 400 \text{ kHz}$$

$$X_L = 2\pi f_R L = 10\text{k}\Omega = X_C$$

When $X_C = X_L$, $|I_C| = |I_L|$

$$I_L = E_S/X_L = 1.4\text{mA} @ -90^\circ$$

$$I_C = E_S/X_C = 1.4\text{mA} @ 90^\circ$$

$$I_T = I_L(\text{phasor}) + I_C(\text{phasor}) = -j1.4\text{mA} + j1.4\text{mA} = 0$$

(so circulating $I_{\text{TANK}} = 1.4\text{mA}$)

Current and Phase Angle at Resonance

- Ideally the total current from the source at resonance is zero because the impedance is infinite.
- In the non-ideal case when the coil resistance is considered, there is some total current at the resonant frequency:

$$I_{\text{tot}} = I_T = I_{\text{Line}} = V_s / Z_r$$

- Since the impedance is purely resistive at resonance, the phase angle is 0° .

Effect of Coil Resistance on the Parallel Resonant Frequency

- Q is the quality factor of the coil, X_L/R_W .
 - There is also a circuit Q (Q_o or Q_{CKT}) which equals $R_{p(tot)}/X_{L(eq)}$. $R_{p(tot)}$ is the equivalent parallel DC resistance of the circuit including R_G , R_{Series} , R_L , $R_{p(eq)}$, and any other resistive branches. More on this later!!!

- For values of $Q_{COIL} \geq 10$, the parallel resonant frequency is:

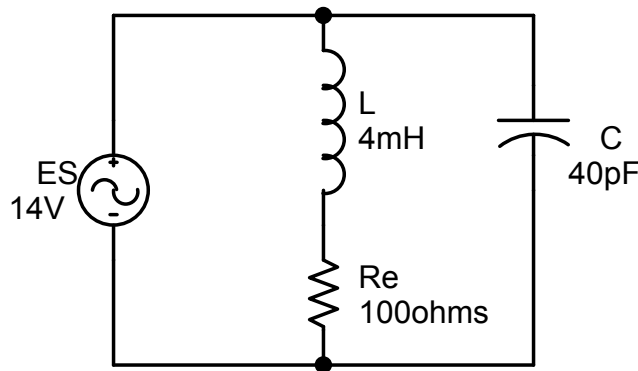
$$f_r \cong 1/(2\pi\sqrt{LC})$$

- If coil $R_{p(tot)} \geq 10R_W(Q_{COIL}^2)$, then Q of coil is essentially Q of circuit. But if $R_{p(tot)} < 10R_W(Q_{COIL}^2)$, then Q of circuit is less than Q of coil.

Practical Parallel Resonant Circuit

Example 1

- Calculate f_R , and X_L , X_C , Z_{COIL} , I_C , I_L , & I_T , & Z_{TANK} at f_R for the practical parallel resonant circuit:



- I_L and I_C will not quite cancel because Z_{COIL} will be slightly greater than X_L due to R_W .
- I_L will not lag E_S by exactly 90° due to R_W , so I_L & I_C will not be quite 180° out of phase.
- At f_R , I_T will be very small if R_W is small compared to X_L , $Q_{\text{COIL}} > 10$, so I_T will be MIN but not zero at f_R .
- Z_{TANK} will be MAX but not ∞ at f_R .

Practical Parallel Resonant Circuit

Example 1 (cont.)

$$f_R = 1/(2\pi\sqrt{LC}) = 400 \text{ kHz}$$

$$X_L = 2\pi f_R L = 10\text{k}\Omega = X_C$$

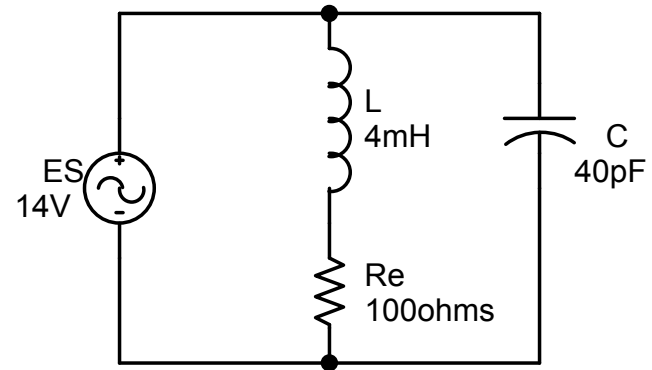
$$I_C = E_S / X_C = 1.4\text{mA} @ 90^\circ = 0 + j1.4\text{mA}$$

$$Z_{\text{COIL}} = R_W + jX_L \text{ (R} \rightarrow \text{P)} = 10\text{k}\Omega @ 89.4^\circ$$

$$I_L = E_S / Z_{\text{COIL}} = 1.4\text{mA} @ -89.4^\circ = 0.014\text{mA} - j1.4\text{mA}$$

$$I_T = 0.014\text{mA} + j0 = 14\mu\text{A} @ 0^\circ$$

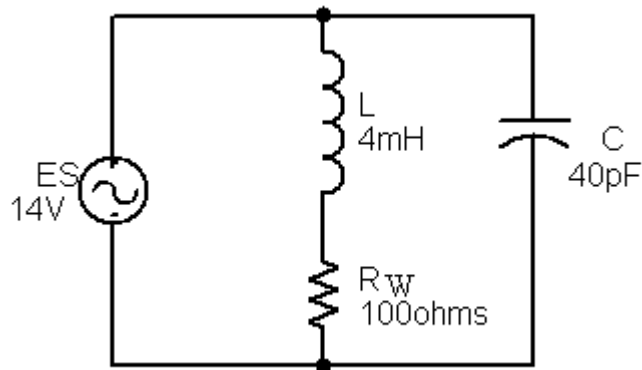
$$Z_{\text{TANK}} = E_S / I_T = 1\text{M}\Omega @ 0^\circ$$



- Let's make a table and compare resonance with frequencies 0.1 above and below f_R

f	I_L	I_C	I_T	Z_T
$0.9f_R = 360\text{kHz}$	1.54mA	1.26mA	0.28mA	$50\text{k}\Omega$ (L)
$f_R = 400\text{kHz}$	1.4mA	1.4mA	0.014mA	$1\text{M}\Omega$ (R)
$1.1f_R = 440\text{kHz}$	1.26mA	1.54mA	0.28mA	$50\text{k}\Omega$ (C)

Current Magnification or Q



f	I _L	I _C	I _T	Z _T
0.9f _R = 360kHz	1.54mA	1.26mA	0.28mA	50kΩ (L)
f _R = 400kHz	1.4mA	1.4mA	0.014mA	1MΩ (R)
1.1f _R = 440kHz	1.26mA	1.54mA	0.28mA	50kΩ (C)

- Note that at $f = f_R$, I_T is only 0.014mA while I_L and I_C are each at 1.4mA. So the Tank Current or circulating current is 100X the source, total, or “line” current. This is called the current magnification factor, or quality factor, or Q.
- $I_L = I_C = Q \times I_T$ or $Q \times I_{\text{LINE}}$ @ $f = f_R$. Remember that I_L or I_C are also called I_{TANK} .
- Also, $Q_{\text{CKT}} = Q_{\text{COIL}} = X_L/R_W = 10\text{k}\Omega/100\Omega = 100$, which is the same as the current magnification. ($Q_{\text{CKT}} = Q_{\text{COIL}}$ assuming $R_G \geq 10R_W(Q_{\text{COIL}}^2)$). Remember, for parallel resonance, $Q_{\text{CKT}} = R_{p(\text{tot})}/X_{L(\text{eq})}$. Since $R_{p(\text{tot})} \approx R_G$ is assumed to be large in this example, $Q_{\text{CKT}} = Q_{\text{COIL}}$).
- Impedance Magnification is also equal to Q, so $Z_{\text{TANK}} = Q \times X_L = Q \times X_C$ @ $f = f_R$.
- For the circuit above, $Z_{\text{TANK}} = 100 \times 10\text{k}\Omega = 1\text{M}\Omega$, the same as previously calculated.

Practical Parallel Resonant Circuit

Example 2

- Given: Parallel Resonant Circuit; $L = 200\mu\text{H}$, $R_W = 60\Omega$, $C = 130\text{pF}$, $E_S = 50\text{mV}$.
- **Find**: f_R , Q , Z_{TANK} , I_{LINE} , I_C , I_L , I_{TANK}
- **Solution**:
 - a) $f_R = 1/(2\pi\sqrt{LC}) = 1 \text{ MHz}$
 - b) $Q = X_L/R_W = (2\pi f_R L)/R_W = 21$ where $X_L = 1.26\text{k}\Omega$
 - c) $Z_{\text{TANK}} = Q \times X_L = 26.5\text{k}\Omega$
 - d) $I_{\text{LINE}} = E_S/Z_{\text{TANK}} = 1.9\mu\text{A}$
 - e) $I_C = I_L = I_{\text{TANK}} = Q \times I_{\text{LINE}} = 40\mu\text{A}$
- **To Summarize so far: For a Parallel Resonant Circuit @ $f = f_R$:**
 - $X_L = X_C$
 - $Z_{\text{TANK}} = \text{MAX}$
 - $I_{\text{LINE}} = \text{MIN}$
 - Z_{TANK} purely resistive
 - I_{LINE} in phase with E_S
 - $Q_{\text{COIL}} = X_L/R_W$

Remember that all the equations are premised on a high Q coil – i.e. at least 10, preferably 20.

Formula for Bandwidth

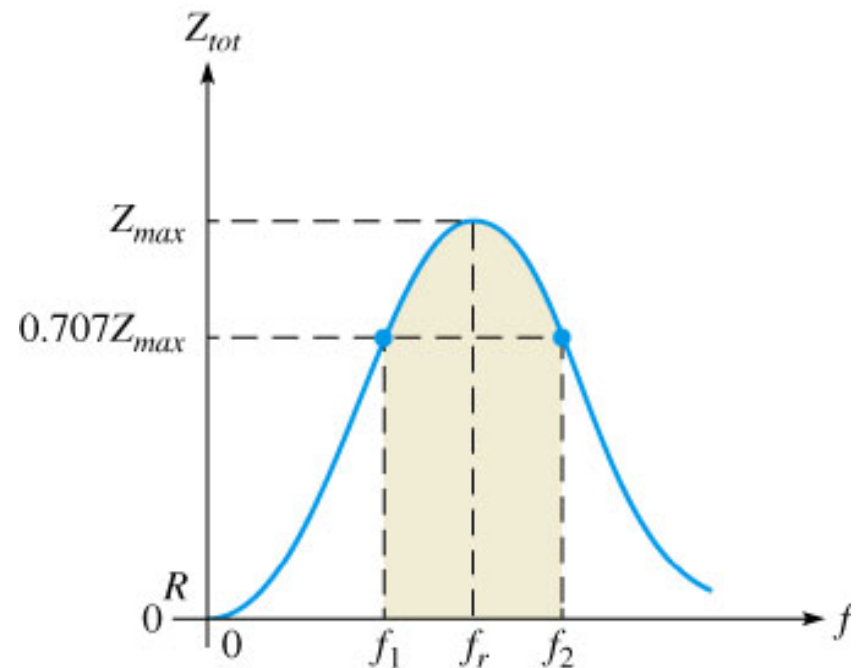
- Bandwidth for parallel resonant circuits is the range of frequencies between the critical frequencies for which the response curve of Z is 0.707 of the maximum value.

$$BW = f_2 - f_1$$

- Ideally the center frequency is:

$$f_r = (f_1 + f_2)/2$$

- BW for AM Radio: 10KHz, 5KHz on each side of the center frequency.
- BW for FM Radio: 200 KHz, 75 KHz on each side of the center frequency plus an additional 25KHz on each side as a guard band.



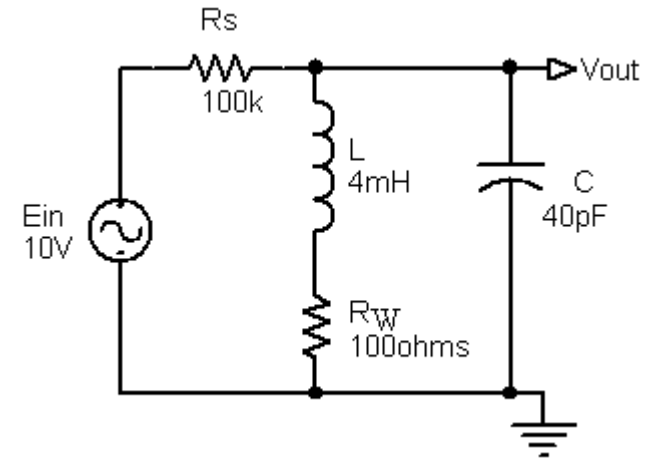
Selectivity and Bandwidth

- The parallel resonant circuit (tank) has $\text{MAX } Z_{\text{TANK}}$ (e.g. $1\text{M}\Omega$) @ f_R . This makes it useful where high impedance is needed over a narrow band of frequencies.
 - i.e. The following could be a Band Pass Filter – note the source impedance in series.

- Since tank impedance is purely resistive at resonance (400kHz) the tank forms a simple resistive voltage divider at resonance.

$$\text{So, } V_{\text{OUT}} = Z_{\text{TANK}} / (Z_{\text{TANK}} + R_S) \times E_{\text{IN}} = 9.1\text{V}$$

- At frequencies above and below resonance, Z_{TANK} decreases rapidly and most of E_{IN} will be dropped across R_S .
- So, V_{OUT} varies as a function of frequency and at or near f_R , there is a large V_{OUT} (max = E_{IN}) – this is called a Bandpass filter.
- Sharpness of response depends on Q .
- $Q_{\text{COIL}} = X_L / R_W$
- $Q_{\text{CKT}} = R_{p(\text{tot})} / X_{L(\text{eq})} = [(R_W(Q_{\text{COIL}}^2 + 1)) || R_S] / X_L$
 - Note, if $R_S \geq 10R_W(Q_{\text{COIL}}^2)$, $Q_{\text{CKT}} = Q_{\text{COIL}} = X_L / R_W$



Example:

Determine BW of tank circuit, and BP (f_1 & f_2)

Solution:

$$Q = X_L / R_W = 100;$$

$$Q_{\text{CKT}} = (1\text{M} || 100\text{k}) / 10\text{k} = 9.1$$

$$\text{BW without } R_S = f_R / Q = 4\text{kHz};$$

$$f_1 = 398\text{kHz},$$

$$f_2 = 402\text{kHz}$$

$$\text{BW with } R_S = f_R / Q = 44\text{kHz};$$

$$f_1 = 378\text{kHz},$$

$$f_2 = 422\text{kHz}$$

OTHER PARALLEL RESONANCE CONCEPTS

- The impedance of the tank and line current depend greatly on inductor winding resistance (R_W).
- The quality factor of the tank depends mainly on the Q of the inductor (X_L/R_W)
- The quality factor of the entire parallel resonant circuit determines the operating bandwidth of the circuit (Q_{CKT})
- The circuit $Q = Q_{CKT}$ is determined by the equivalent parallel resistance and tank impedance. The higher the parallel resistance and tank impedance, the higher the circuit Q.
- Resistance in series with the tank should be considered as a parallel resistance for the purpose of circuit Q calculations.
- The generator resistance cannot be ignored if the analysis of a parallel resonant circuit is to be accurate. Ideally, the generator resistance should be infinite for the highest possible circuit Q.
- Changing L or C will change the resonant frequency of a tank. Changing R_p or R_W will change the circuit Q and bandwidth.

Conversion of Series-Parallel to Parallel

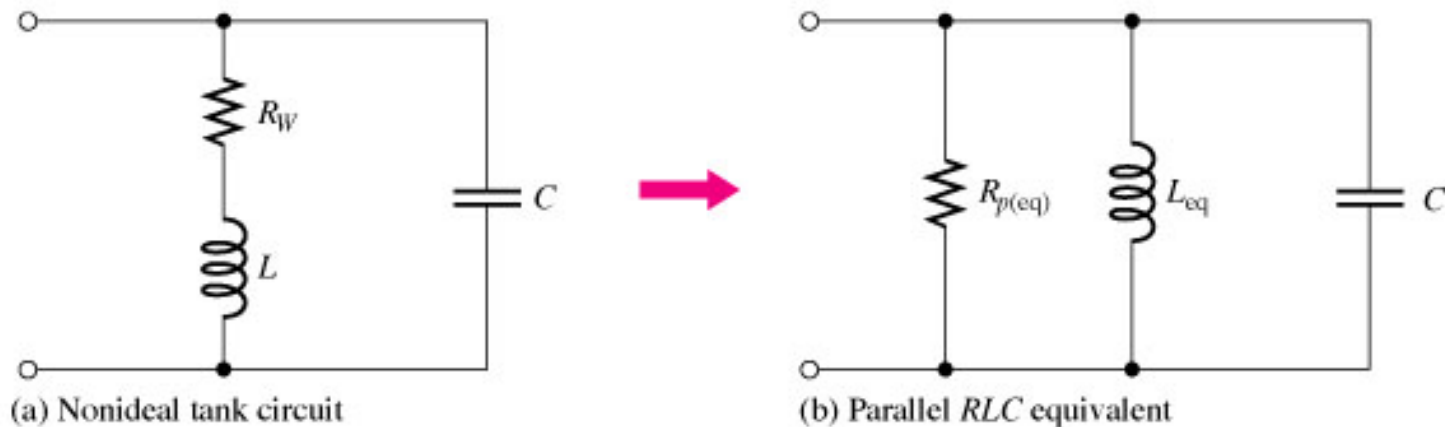
- Equivalent Inductance:

$$L_{eq} = L((Q^2+1)/Q^2)$$
$$L_{eq} \approx L \text{ if } Q \geq 10$$

- Equivalent Resistance:

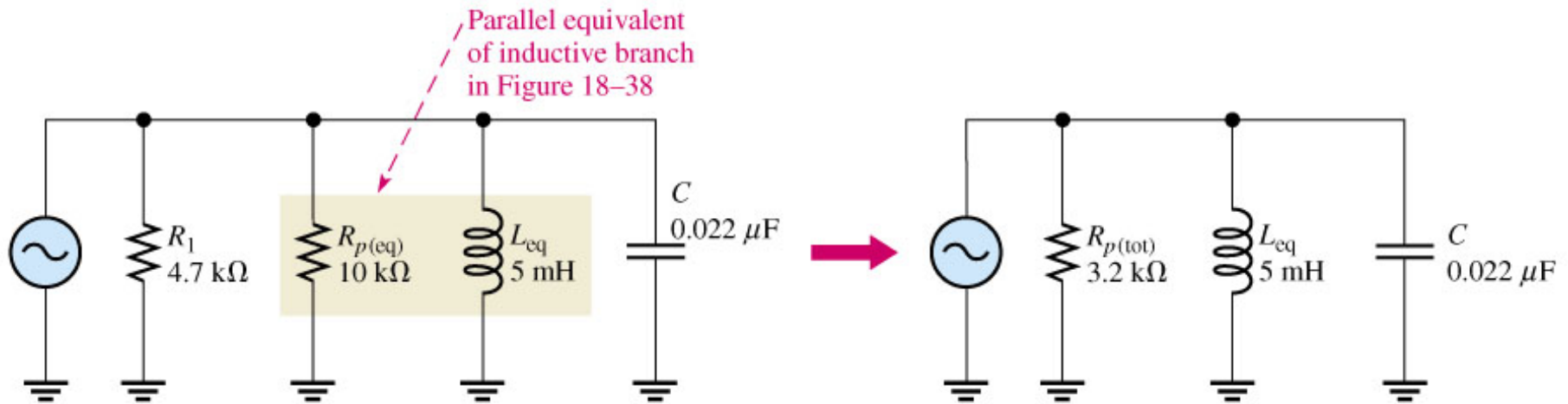
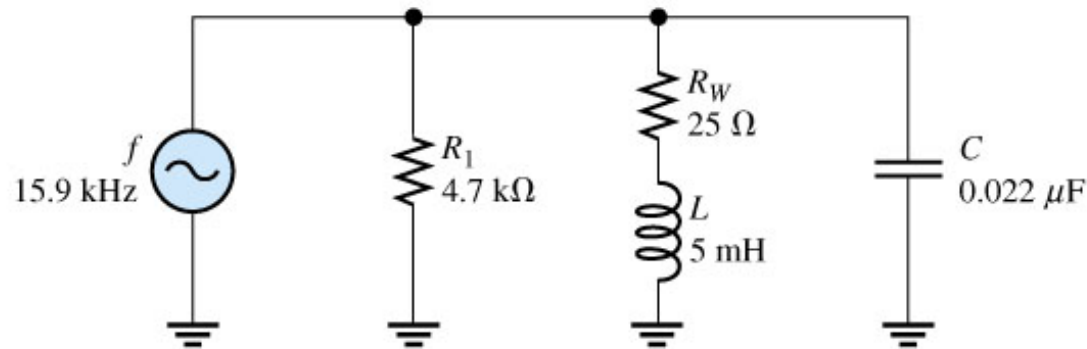
$$R_{p(eq)} = Z_{TANK(unloaded)} = R_W(Q^2+1) \approx X_L^2/R_W \text{ or } X_C^2/R_W$$

derivation: $R_W(X_L/R_W)^2 + R_W$ (drop the $+R_W$)



- Sometimes, makes circuit analysis easier

Series-Parallel Conversion Example



(a) Parallel equivalent of the circuit in Figure 18-38

(b) $R_{p(\text{tot})} = R_1 \parallel R_{p(\text{eq})} = 3.2 \text{ k}\Omega$

- $X_L = 500 \text{ ohms}$, $Q = 50$, $R_{p(\text{eq})} = 25 \text{ ohms}(50^2 + 1) = 10 \text{ kohms}$

Parallel Resonant Conditions in a Non-ideal Circuit

- A practical treatment of parallel resonant circuits must include the coil resistance.
- At parallel resonance:

$$X_{L(\text{eq})} = X_C$$

- The total impedance of the non-ideal tank circuit at resonance can be expressed as the equivalent parallel resistance:

$$Z_r = Z_{\text{TANK}(\text{unloaded})} = R_W(Q^2 + 1)$$

- Note: $Q \geq 10$

Figure 18--40 A practical treatment of parallel resonant circuits must include the coil resistance.

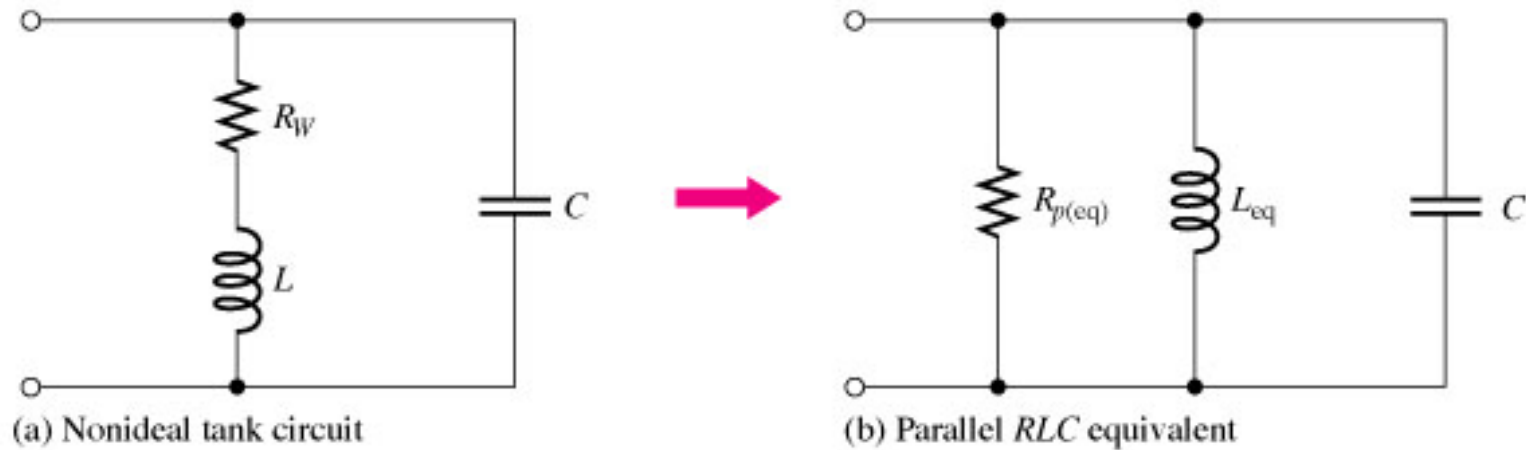
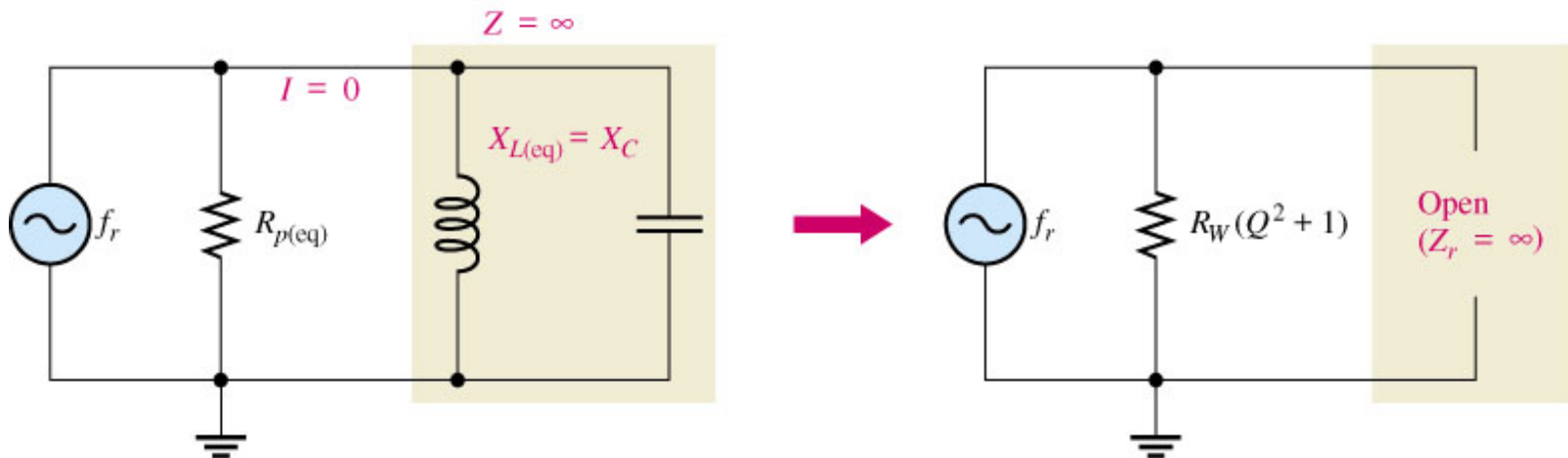


Figure 18--41 At resonance, the parallel LC portion appears open and the source sees only $R_{p(eq)}$.



External Parallel Load Resistance

- Often an external load resistance appears in parallel with a tank circuit.

- The external resistance effectively appears in parallel with the equivalent parallel resistance ($R_{p(eq)}$) of the coil:

$$R_{p(tot)} = Z_{TANK(loaded)} = R_L || R_{p(eq)}$$

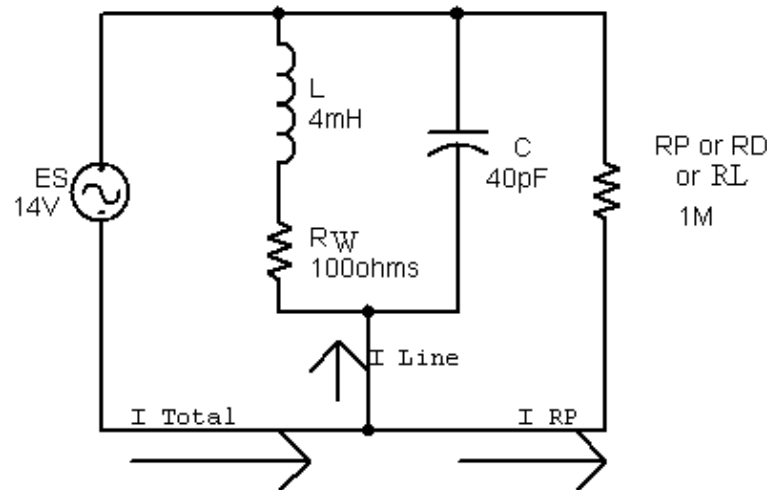
- The external resistor (R_L) will lower the overall Q, designated Q_O of the circuit:

$$Q_O = R_{p(tot)} / X_{L(eq)}$$

- This effect is called “damping” and is a common method used to increase the bandwidth of a parallel resonant circuit at the expense of reduced selectivity.

Damping Example

- If a resistor is connected in parallel (shunt) with the TANK, it can reduce Q and increase BW. This is called DAMPING.



- $Q_{(\text{unloaded})} = 100$

- For the demonstration tank circuit above, $f_R = 400\text{kHz}$, $Z_{\text{TANK}(\text{unloaded})} = 1\text{M}\Omega$, $I_{\text{LINE}} = 14\mu\text{A}$ without the damping resistor RP or RD or RL.
- If a damping resistor is added that is equal to the tank impedance at resonance, the total impedance of the damped tank is: $Z_T = Z_{\text{TANK}} \parallel R_P$
- $I_T = I_{\text{LINE}} + I_{\text{RP}}$ where $I_{\text{RP}} = E_S/R_P = 14\mu\text{A}$, and $I_T = 14\mu\text{A} + 14\mu\text{A} = 28\mu\text{A}$
But also $I_T = E_S/Z_T = 28\mu\text{A}$
- The Damped $Q = R_{p(\text{tot})}/X_L = [(X_L^2/R_W) \parallel R_P]/X_L = [((10\text{k})^2/100\Omega) \parallel 1\text{M}]/10\text{k} = \underline{50}$
- The Damped $Q = \frac{X_L}{R_W + (X_L^2/R_P)} = \underline{50}$
- $\text{BW} = f_r/Q = 400\text{KHz}/50 = \underline{8\text{KHz}}$ Without Damping Resistor, $Q = 100$, $\text{BW} = \underline{4\text{KHz}}$

Figure 18--49 A basic tuned band-pass amplifier.

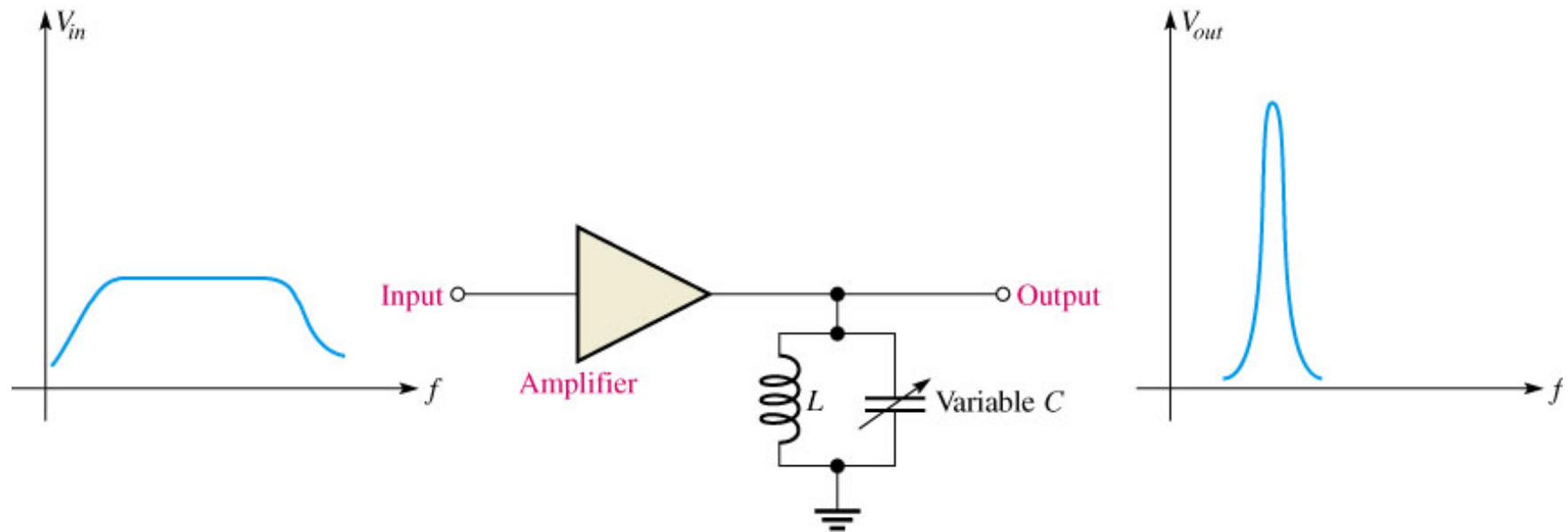


Figure 18--50 Resonant coupling from an antenna.

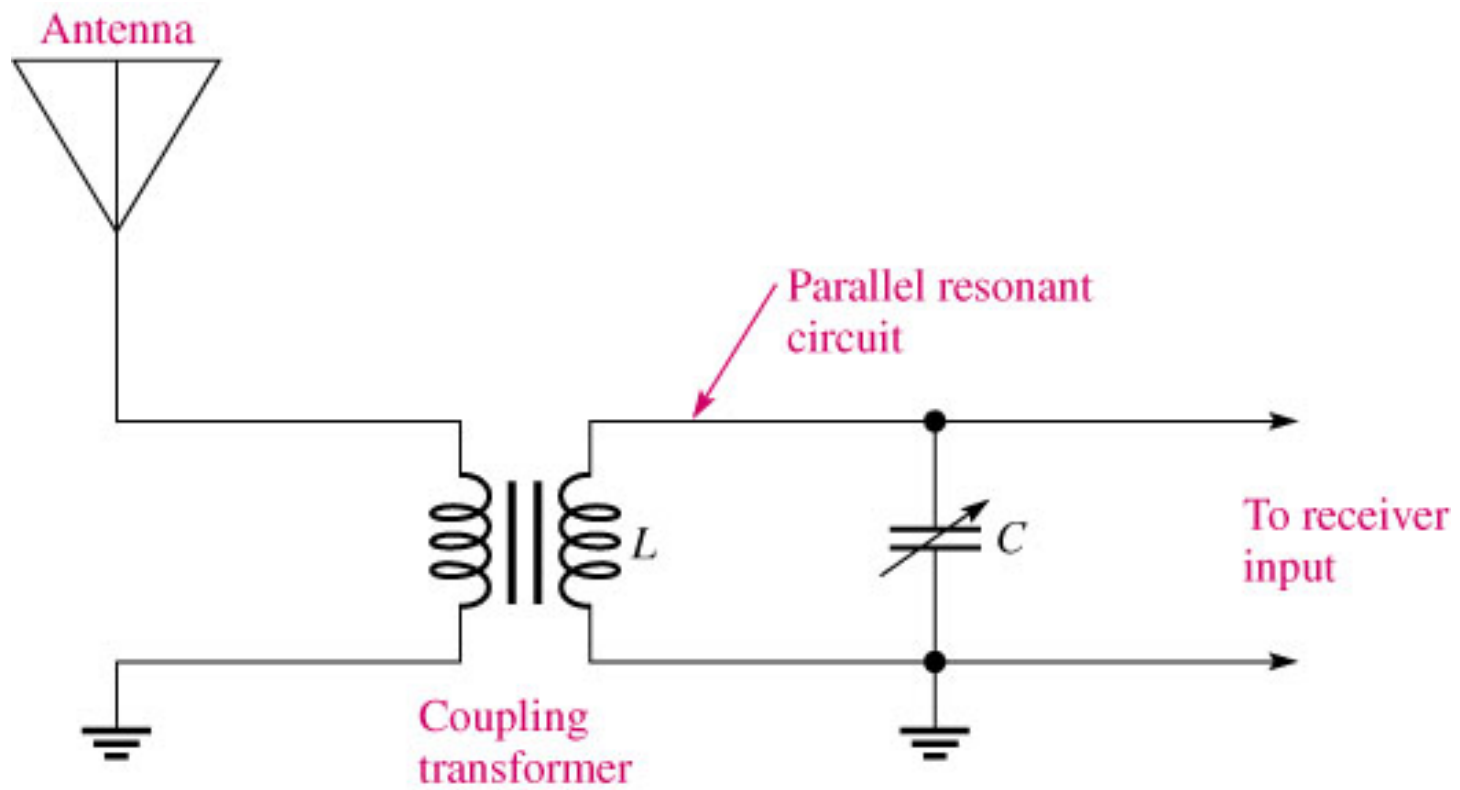


Figure 18--51 Double-tuned amplifiers.

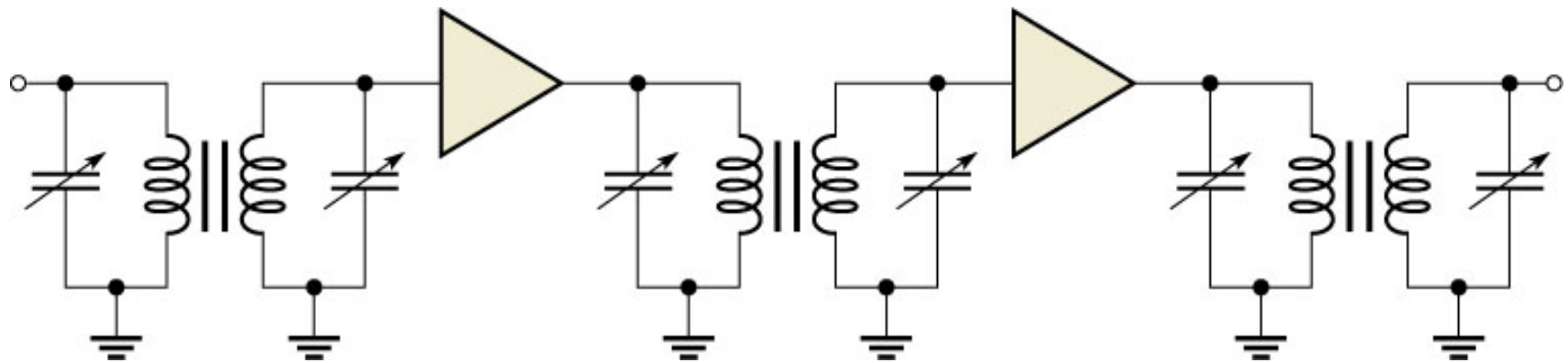


Figure 18--52 A simplified portion of a TV receiver showing filter usage.

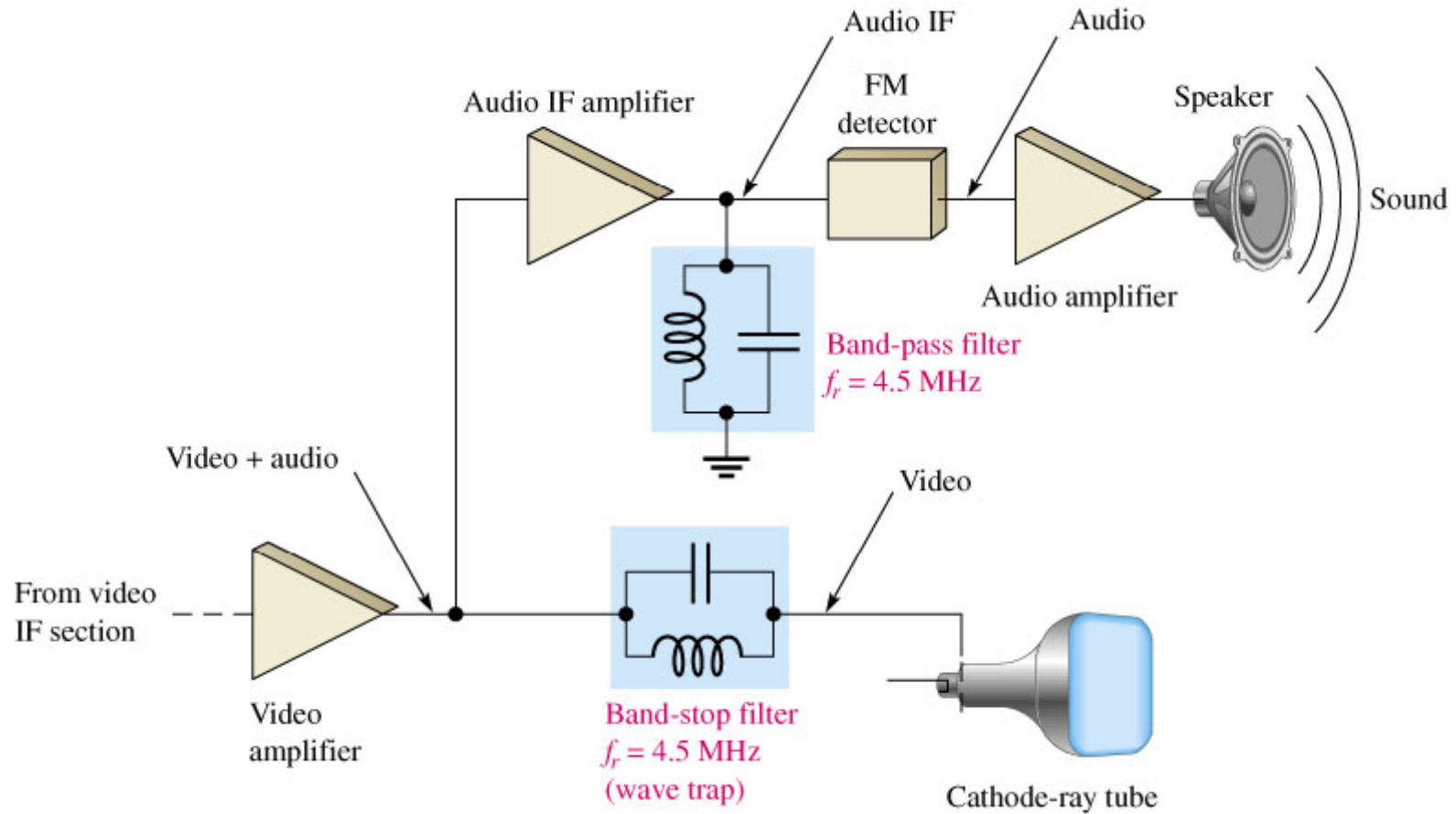


Figure 18–53 A simplified diagram of a superheterodyne AM radio broadcast receiver showing an example of the application of tuned resonant circuits.

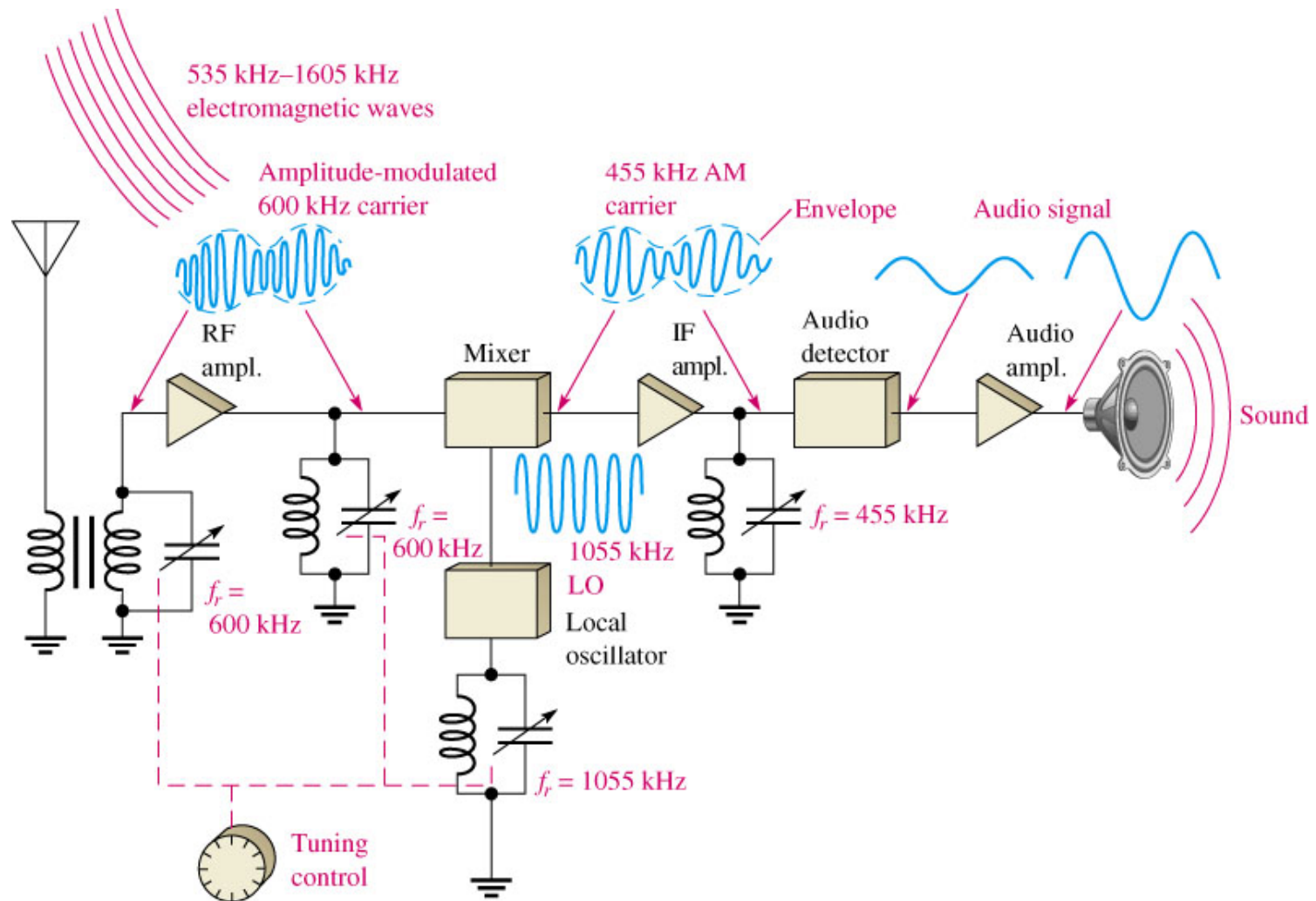


Figure 18--54 Simplified block diagram of a basic radio receiver.

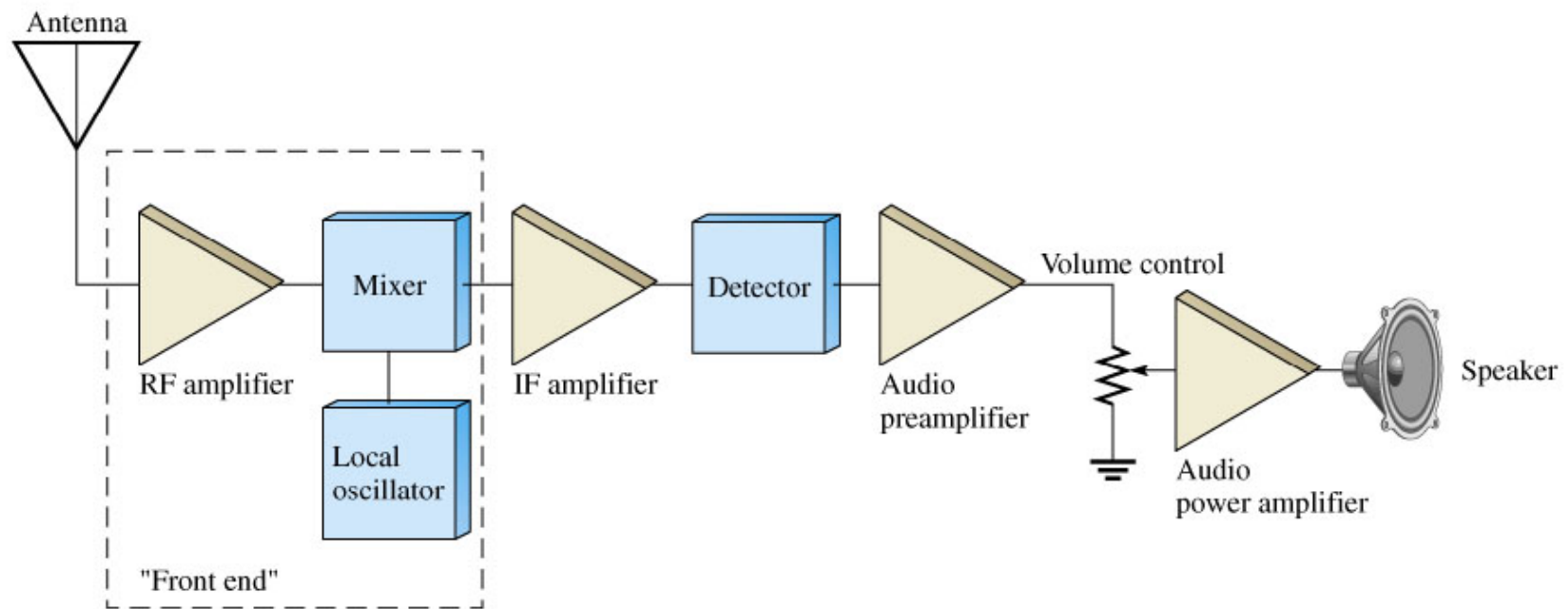
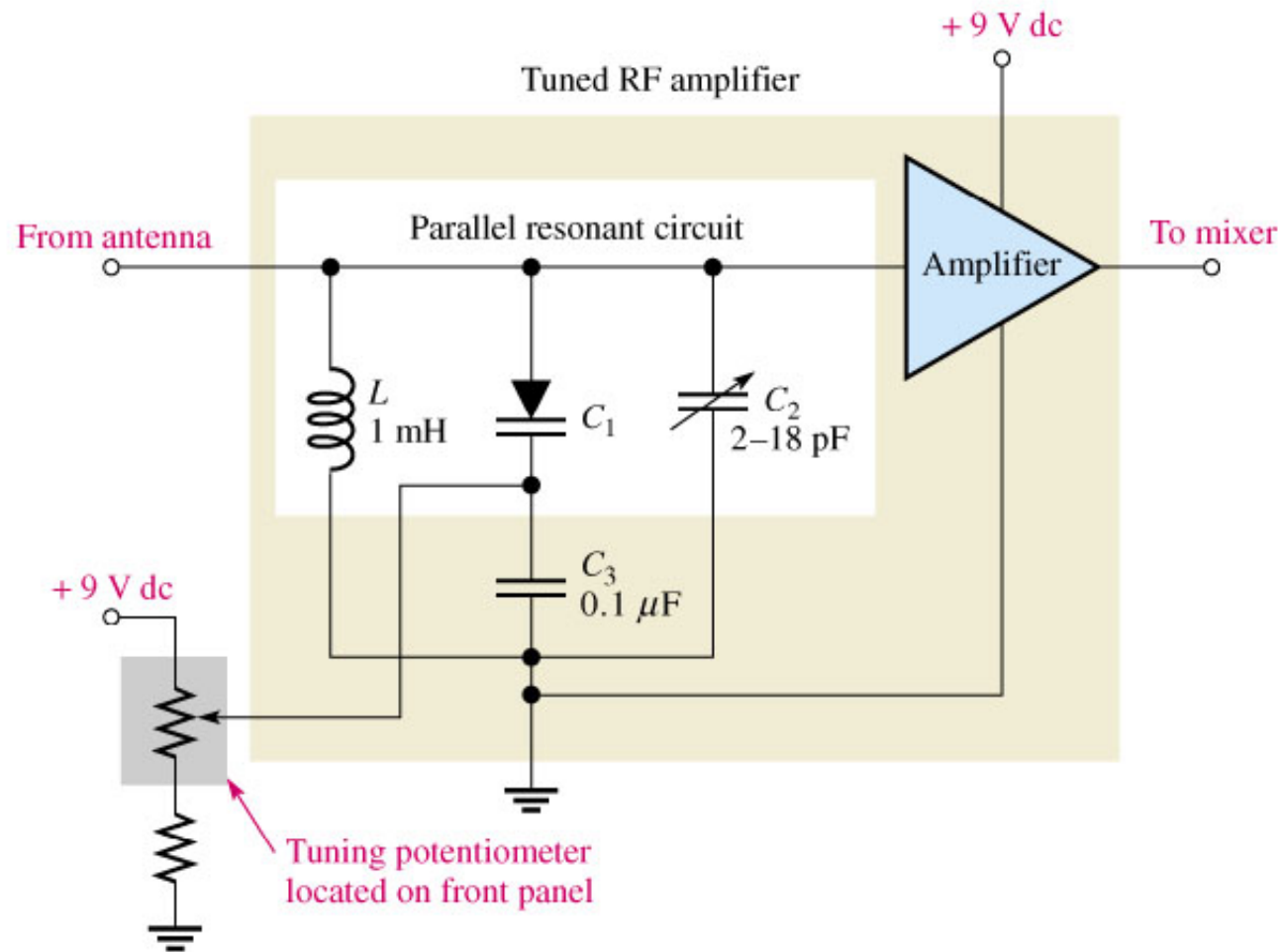


Figure 18--55 Partial schematic of the RF amplifier showing the resonant tuning circuit.



Summary

- X_L and X_C have opposing effects in a RLC circuit.
- In a series RLC circuit, the larger reactance determines the net reactance of the circuit.
- At series resonance, the inductive and capacitive reactances are equal.
- The impedance of a series RLC circuit is purely resistive at resonance.
- In a series RLC circuit, the current is maximum at resonance.

Summary

- The reactive voltages V_L and V_C cancel at resonance in a series RLC circuit because they are equal in magnitude and 180° out of phase.
- In a parallel RLC circuit, the smaller reactance determine the net reactance of the circuit.
- In a parallel resonant circuit, the the impedance is maximum at the resonant frequency.
- A parallel resonant circuit is commonly called a **tank circuit**.

Summary

- The impedance of a parallel RLC circuit is purely resistive at resonance.
- The bandwidth of a series resonant circuit is the range of frequencies for which the current is $0.707I_{\max}$ or greater.
- The bandwidth of a parallel resonant circuit is the range of frequencies for which the impedance is $0.707Z_{\max}$ or greater.

Summary

- The critical frequencies are the frequencies above and below resonance where the circuit response is 70.7% of the maximum response.
- A higher Q produces a narrower bandwidth.